COMPUTATIONAL GAME THEORY

Exercises

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October 27, 2026

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ABOUT THIS DOCUMENT

The purpose of these exercises is to help you, as students of *Computational Game Theory*, actively engage with the concepts introduced in the lectures. You are encouraged to develop your own approaches to solving the problems and to use the exercises as an opportunity to deepen your understanding of the material. Most exercises can be solved with pen and paper, and you are welcome to use the wide margins of this document for notes, calculations, and sketches of arguments.

When needed, consult the online course materials for the relevant definitions, fundamental results, and examples. Many of the exercises are adapted from the recommended textbooks [1, 2, 3, 4]. Some questions are more demanding and require longer mathematical arguments; these are marked with ★ .

In addition to exercises directly related to the lecture content, the first section recalls selected mathematical prerequisites needed for the study of game theory. This section is based primarily on material from undergraduate courses in linear algebra, optimization, linear programming, discrete mathematics, and probability theory.

1 MATHEMATICAL PREREQUISITIES

Exercise 1.1

Duality in linear programming. The concept of duality appears in many game-theoretic problems. In this exercise we consider the following linear program:

$$\begin{aligned} \text{Minimize} \quad & -x_1 + 2x_2 \\ \text{subject to} \quad & x_1 + x_2 \leq 4 \\ & 2x_1 + x_2 \leq 5 \\ & -x_1 + 4x_2 \geq 2 \\ & x_1, x_2 \geq 0 \end{aligned}$$

Characterize the feasible set of the problem by its vertices (extreme points). Additionally, find the optimal solution of this linear program and formulate the dual program. What is the meaning of the dual program and its relation to the original program? What will be the optimal value of the dual problem?

Exercise 1.2

Minimax/maximin values. Let X and Y be nonempty sets and consider a real function $f\colon X \times Y \to \mathbb{R}$. Show that

$$\max_{x \in X} \min_{y \in Y} f(x,y) \leq \min_{y \in Y} \max_{x \in X} f(x,y), \tag{1}$$

where we assume that all maxima and minima above exist. Find an example of function f taking at least two different values such that (a) the inequality is strict and (b) the inequality becomes an equality. *Hint:* Such examples exist already when both X and Y have 2 elements.

Exercise 1.3

Saddle points. Let $f\colon X \times Y \to \mathbb{R}$ be a function, where X and Y are arbitrary nonempty set. We say that $(x^*, y^*) \in X \times Y$ is a *saddle point* of f if

$$f(x, y^*) \leq f(x^*, y^*) \leq f(x^*, y) \qquad \text{for all } x \in X,\ y \in Y. \tag{2}$$

Show that a function f has a saddle point if, and only if,

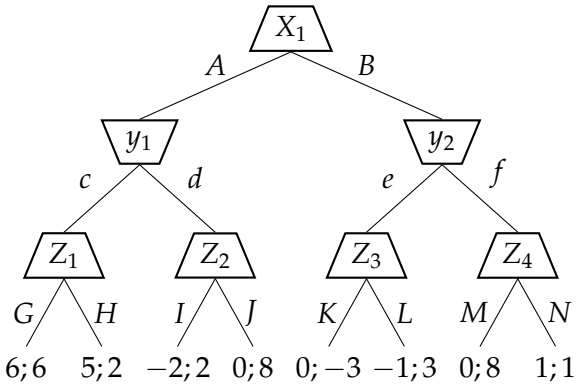
$$\max_{x \in X} \min_{y \in Y} f(x,y) = \min_{y \in Y} \max_{x \in X} f(x,y), \tag{3}$$

provided that all the maxima/minima above exist.

Exercise 1.4

Consider a game called Nim played by two players. There are 21 matches on board. The players take turns removing 1, 2, or 3 matches from the board. The player that takes the last match loses. What is the strategy you should play in such a game?

Exercise 1.5
Consider following sequential game played by 2 players. What is the rational strategy of both players in such a game, if player 1 acts first and if player 2 acts first.



Exercise 1.6
Let $N = \{1, \dots, n\}$ and fix a player $i \in N$. A coalition $A \subseteq N \setminus \{i\}$ is drawn at random. Compute the probability of obtaining a given coalition A under each of the following processes.

(a) A is drawn uniformly from all subsets of $N \setminus \{i\}$.

(b) A size a is drawn uniformly from $\{0, 1, \dots, n-1\}$, and then A is drawn uniformly from the subsets of $N \setminus \{i\}$ of size a.

(c) The n players are arranged in a uniformly random order, and A is the set of players preceding i.

SOLUTIONS

Solution 1.1

In any linear programming problem, the feasible set is a convex polyhedron. In our case the polyhedron is bounded, so it is a convex polygon characterized by its vertices. Every vertex corresponds to a unique solution of some linear equality system associated with a subset of linear inequality constraints of the problem. For example, a point $(1,3)$ is a vertex since it is a unique solution to linear equations $x_1 + x_2 = 4$ and $2x_1 + x_2 = 5$. The remaining vertices are $(0, \frac{1}{2}), (0,4), (2,1)$. See Figure 1, which is an output of the simple online solver `https://online-optimizer.appspot.com`.

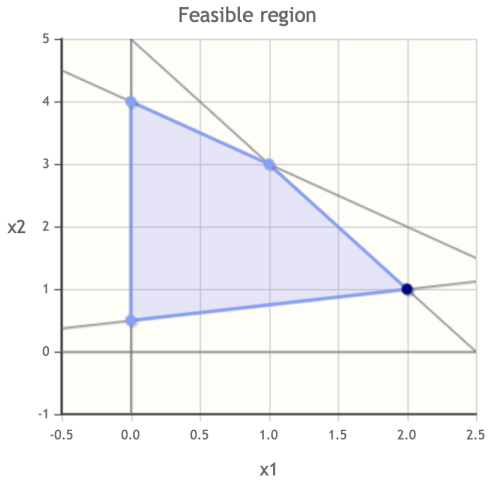


Figure 1: The feasible set corresponding to constraints (5)–(7)

What is the optimal solution to the problem? The fundamental theorem of linear programming says that, if a linear program has an optimal solution, then it is attained at some vertex of the feasible set. Clearly, at least one optimal solution exists in our case since the feasible set is bounded. Therefore, it suffices to compute the values of the objective function $f(x_1, x_2) = -x_1 + 2x_2$ for all the vertices and select the one with the minimal value. This yields the optimal solution $(2,1)$ with the optimal value is $f(2,1) = 0$.

To formulate the dual problem, it is convenient to formulate the original problem so that all inequalities are of the form $\geq$:

$$\begin{aligned} \text{Minimize} \quad & -x_1 + 2x_2 && (4)\\ \text{subject to} \quad & -x_1 - x_2 \geq -4 && (5)\\ & -2x_1 - x_2 \geq -5 && (6)\\ & -x_1 + 4x_2 \geq 2 && (7)\\ & x_1, x_2 \geq 0 && (8) \end{aligned}$$

By design, the dual program has 3 variables y_1, y_2, y_3 corresponding to the 3 primal linear constraints (5)–(7), and 2 linear constraints associated with

2 primal variables x_1, x_2. The exact form of the dual program can be found in any textbook on linear programming. Specifically the dual is

$$\begin{aligned} \text{Maximize} \quad & -4y_1 - 5y_2 + 2y_3 \\ \text{subject to} \quad & -y_1 - 2y_2 - y_3 \leq -1 \\ & -y_1 - y_2 + 4y_3 \leq 2 \\ & y_1, y_2, y_3 \geq 0 \end{aligned}$$

The duality theorem immediately implies that the primal and dual problems have the same optimal value 0.

What is the interpretation of duality? The optimal value of dual problem provides the tighest lower bound on the solution of the primal problem. In fact, the dual problem controls the optimal value of the primal problem from below, and does so in the best possible way — the optimal values of both programs coincide!

We will elaborate on the idea sketched above. How to obtain a lower bound for the primal problem? Note that one possible lower bound on the values of the objective f follows from the nonnegativity constraints (8) and the constraint (5) since

$$-4 \leq -x_1 - x_2 \leq -x_1 + 2x_2.$$

Can we do better? That is, can we obtain a higher value than -4 for the lower bound? It is easy to see that by multiplying the inequalities (5) and (7) by $\frac{1}{2}$ and adding them we obtain the bound

$$-1 \leq -x_1 + \tfrac{3}{2}x_2 \leq -x_1 + 2x_2,$$

which is tighter. Can we still improve on this lower bound? Let us try to generalize the idea of combining the linear inequalities of the primal problem for estimating the optimal value from below. The goal is to find nonnegative real numbers y_1, y_2, y_3 such that

$$\begin{aligned} -4y_1 - 5y_2 + 2y_3 &\leq y_1(-x_1 - x_2) + y_2(-2x_1 - x_2) + y_3(-x_1 + 4x_2) \\ &= (-y_1 - 2y_2 - y_3)x_1 + (-y_1 - y_2 + 4y_3)x_2 \\ &\leq -x_1 + 2x_2 \end{aligned}$$

and the second inequality should be as tight as possible. One way to guarantee this is to maximize the most left-hand side $-4y_1 - 5y_2 + 2y_3$ while preserving the constraints $-y_1 - 2y_2 - y_3 \leq -1$ and $-y_1 - y_2 + 4y_3 \leq 2$ expressed by the second inequality above. In other words, we have derived precisely the dual problem.

Solution 1.2

To prove the inequality (1), define

$$F(x) = \min_{y \in Y} f(x, y), \qquad x \in X,$$

and let $x^* \in X$ be the maximizer of F over X. Then

$$\max_{x \in X} \min_{y \in Y} f(x, y) = \max_{x \in X} F(x) = F(x^*) = \min_{y \in Y} f(x^*, y).$$

For every $y \in Y$, we have

$$f(x^*, y) \leq \max_{x \in X} f(x, y)$$

by the definition of maximum, which implies

$$\min_{y \in Y} f(x^*, y) \leq \min_{y \in Y} \max_{x \in X} f(x, y).$$

Hence, (1) is proved.

We will construct two examples (a)-(b) under the assumption that $X = Y = \{1, 2\}$. First, consider a function f given by the matrix

X \ Y	1	2
1	0	1
2	1	0

Then

$$\max_{x \in X} \min_{y \in Y} f(x, y) = 0 < 1 = \min_{y \in Y} \max_{x \in X} f(x, y).$$

For the second example (b), take a matrix

X \ Y	1	2
1	3	0
2	2	2

Then

$$\max_{x \in X} \min_{y \in Y} f(x, y) = 2 = \min_{y \in Y} \max_{x \in X} f(x, y).$$

Solution 1.3

We define two functions: $F(x) = \min_{y \in Y} f(x, y)$, for all $x \in X$, and $G(y) = \max_{x \in X} f(x, y)$, for every $y \in Y$. Assume that f has a saddle point (x^*, y^*). Then the saddle point definition (2) and the definition of maxima/minima imply

$$\begin{aligned} \min_{y \in Y} G(y) \leq G(y^*) = \max_{x \in X} f(x, y^*) = f(x^*, y^*) \\ = \min_{y \in Y} f(x^*, y) = F(x^*) \leq \max_{x \in X} F(x). \end{aligned}$$

We obtained the inequality

$$\min_{y \in Y} \max_{x \in X} f(x, y) \leq \max_{x \in X} \min_{y \in Y} f(x, y),$$

and since the converse inequality always holds (Exercise 2), we proved the minimax equality (3).

To prove the converse implication, assume that (3) holds. Let x^* and y^* be such that $F(x^*) = \max_{x \in X} F(x)$ and $G(y^*) = \min_{y \in Y} G(y)$, respectively. Then the assumption (3) gives

$$\begin{aligned} F(x^*) = \max_{x \in X} F(x) = \max_{x \in X} \min_{y \in Y} f(x, y) \\ = \min_{y \in Y} \max_{x \in X} f(x, y) = \min_{y \in Y} G(y) = G(y^*). \end{aligned} \tag{9}$$

Further, from the definition of F, G and x^*, y^* we get

$$F(x^*) \leq f(x^*, y^*) \leq G(y^*)$$

and (9) implies that even $F(x^*) = f(x^*, y^*) = G(y^*)$. Let $x \in X$ and $y \in Y$. Then

$$f(x, y^*) \leq G(y^*) = f(x^*, y^*) = F(x^*) \leq f(x^*, y).$$

Solution 1.4

Instead of asking what you should do when there are 21 matches, let us start with an easier version of this problem: There is only 1 match. In that case, the first player takes it and immediately loses, so the winner is player 2 in such a game. When there are 2 matches, the optimal strategy is to remove a single match and force player 2 to take the last one. The same logic can be applied for 3 and 4 matches. In those cases, the player 1 is the winner. Now, when there are 5 matches, player 1 is forced to take 1,2 or 3 matches, which results in 2, 3, or 4 matches on board, respectively. The player is destined to lose in each of those if the opponent plays perfectly. So, no matter how many matches it takes, it loses.

Now, let us generalize this idea. There are a certain number of matches with which the player always loses against an optimal opponent. Let us call those losing amounts. We have seen that those amounts are 1 or 5. On the other hand, there are amounts from which the player can remove a certain amount of matches to end up with a losing amount for the opponent. We have seen that those are 2, 3 and 4. Let us call them winning amounts.

Since we know 5 is a losing amount from the player's perspective, then it is rational from the player's perspective to force the opponent to this losing amount. This can be achieved from 6,7,8 amounts. However, with 9 matches, the player cannot force the opponent into this losing state. Similar logic can be applied further ending up with losing amounts being

$$4n + 1 \quad \forall n \in \mathcal{N}$$

All the others are winning amounts because, from those, you will force the opponent into the losing amount.

The optimal strategy when there is a losing amount is to take an arbitrary amount of matches since you cannot win against a rational opponent. With the winning amount, the optimal strategy is always to take set amount of matches to end up in the losing amount.

Consider an algorithm called backward induction. This algorithm follows a similar approach as the one describes above. it starts at the end of the game tree (terminal states) and based on actions it goes backwards to the root (initial state) of the game.

Solution 1.5

Performing backward induction finds rational strategies. If player 1 is root, it also acts in Z_i. Since rational player is trying to maximize then in Z_1, it will choose action G, because $6 > 5$. Similarly in Z_2 it will choose J because $-2 < 0$. Now the values of Z_1 and Z_2 are $6;6$ and $0;8$ respectively. So in y_1, the rational player picks d, because $6 < 8$, which yields value of y_1 to $0;8$.

Similarly in the right subtree the rational player chooses K or N. This yields values for Z_3 and Z_4 to be $0; -3$ and $1; 1$ respectively. In y_2 it is then rational to choose f, so the value is $1; 1$. Lastly in X_1 the optimal action is B, because $0 < 1$. If both players play rationally, they will both receive reward 1.

When swapping the player the procedure is similar. Now in last decision nodes the player plays G, J, K, M, so the values are

$$Z_1 \to 6; 6 \tag{10}$$
$$Z_2 \to 0; 8 \tag{11}$$
$$Z_3 \to -1; 3 \tag{12}$$
$$Z_4 \to 0; 8 \tag{13}$$

The rational response for player 1 is to pick c and f

$$y_1 \to 6; 6 \tag{14}$$
$$y_2 \to 0; 8 \tag{15}$$

In the end the second player chooses action B because $6 < 8$. So the resulting reward would be 0 for player 1 and 8 for player 2.

Solution 1.6

(a) There are 2^{n-1} coalitions A to which player i doesn't belong, $i \notin A$. Therefore $p(A) = 2^{1-n}$ is the probability of picking such a coalition randomly. Consider now the second variant of random choice of the coalition. Let A be a random coalition selected in this way. We can compute the probability $q(A)$ using the law of total probability,

(b) The process produces A precisely when it draws the size $|A|$, which happens with probability $1/n$, and then selects A among the $\binom{n-1}{|A|}$ coalitions of that size, which happens with probability $1/\binom{n-1}{|A|}$. Hence

$$q(A) = \frac{1}{n} \cdot \frac{1}{\binom{n-1}{|A|}}.$$

This is a probability distribution:

$$\sum_{A \subseteq N \setminus \{i\}} q(A) = \frac{1}{n} \sum_{a=0}^{n-1} \binom{n-1}{a} \frac{1}{\binom{n-1}{a}} = \frac{1}{n} \cdot n = 1.$$

(b) = (c). Expanding the binomial coefficient in (b) and using $n\,(n-1)! = n!$,

$$\frac{1}{n\binom{n-1}{a}} = \frac{1}{n} \cdot \frac{a!\,(n-1-a)!}{(n-1)!} = \frac{a!\,(n-1-a)!}{n!},$$

which is exactly the expression obtained in (c). The two mechanisms therefore induce the same distribution. This is the weighting that appears in the Shapley value; the uniform distribution p of (a) is the one appearing in the Banzhaf value. Note that both p and q depend only on the size of each coalition. Specifically, if A and B are coalitions such that $|A| = |B|$, then $p(A) = p(B) = 2^{1-n}$ and $q(A) = q(B)$. Probability distributions p and q appear in the Banzhaf and Shapley values in coalitional game theory. Observe that there is a several order magnitude difference between them (Figure 2). For example, in case that there are $n = 60$ players, the probabilities

of coalition A with $|A| = 10$ are $p(A) \approx 1.7 \times 10^{-18}$ and $q(A) \approx 2.6 \times 10^{-13}$, respectively.

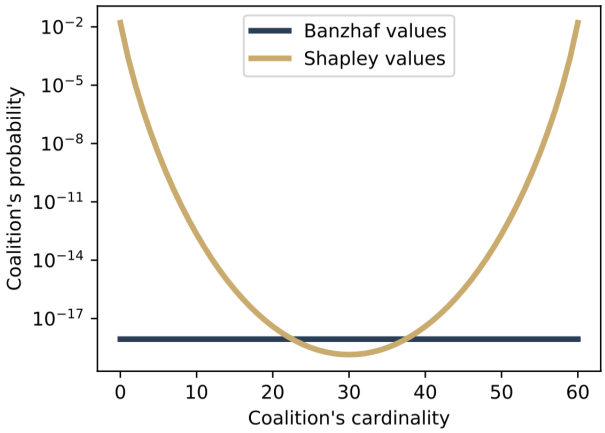


Figure 2: Values of p and q.

2 NORMAL-FORM GAMES

Exercise 2.7

Single attacker is about to attack one of the 4 military bases called $\mathbf{T}_1$, $\mathbf{T}_2$, $\mathbf{T}_3$, and $\mathbf{T}_4$. However, due to the budget restriction, the military can defend only one of the bases. If the attacker attacks the defended base, it gets a utility of -1, while the military gets 2. On the other hand, when the attack is succesful, the military does not receive anything, but the attacker gets the following rewards based on the base it attacked $\mathbf{T}_1 \to 3$, $\mathbf{T}_2 \to 7$, $\mathbf{T}_3 \to 1$, and $\mathbf{T}_4 \to 5$. (a) Formalize this situation as a normal-form game. (b) Let us assume that when the attacker attacks base, the alarm is triggered. If the attacked base is near the protected one, the military may dispatch striking team, which has 50 % chance to interrupt the attack. For base $\mathbf{T}_i$, the nearby bases are $\mathbf{T}_{i-1}$ and $\mathbf{T}_{i+1}$. Bases $\mathbf{T}_1$ and $\mathbf{T}_4$ are also close to each other. (c) Find all pure Nash equilibria in games (a) and (b).

Exercise 2.8

Is there a game with at least one pure Nash equilibrium which is removed during the process of iterated elimination? Either show that such a game cannot exist or find an example that it does.

Exercise 2.9

Consider a two-player normal-form game with the payoff matrix

		Player 2		
		V	**S**	**R**
Player 1	**U**	$1,3$	$4,2$	$-1,2$
	C	$1,0$	$2,-2$	$0,-1$
	D	$1,2$	$-1,1$	$3,3$

(a) Find all Pareto optimal outcomes. (b) Find all pure Nash equilibria. (c) Find all dominated pure strategies and apply the procedure of iterative removal of dominated strategies to this game.

Exercise 2.10

We present the game called *Battle of the Sexes*. Its name is derived from the situation where a couple (Alice and Bob) is trying to plan what to do on Saturday. The alternatives are going to a concert (C) or watching a football match (F). Bob prefers football and Alice prefers the concert, but both prefer being together to being alone, even if that means agreeing to the less-preferred recreational activity.

		Bob	
		C	*F*
Alice	*C*	$2,1$	$0,0$
	F	$0,0$	$1,2$

Find equilibrium strategies of this game.

Exercise 2.11

We present the game called *Game of chicken*. Two drivers race towards each other on a narrow road. Each may *swerve* (S) or *drive straight* (D). A driver who swerves while the other drives straight loses face; a driver who drives straight while the other swerves wins the contest. If both swerve, nothing happens; if neither swerves, they crash.

		Driver 2	
		S	D
Driver 1	S	$0, 0$	$-1, 1$
	D	$1, -1$	$-10, -10$

(a) Find all Nash equilibrium strategies of this game.

(b) Driver 1 is riding a motorcycle and Driver 2 an armoured car, so a crash costs them -10 and -5 respectively. Before computing: which driver do you expect to drive straight more often? Now find the mixed equilibrium.

($\star$) (c) Let the crash cost driver i the amount $-c_i$, where $c_i > 1$. Find the mixed equilibrium and the equilibrium payoffs.

SOLUTIONS

Solution 2.7

(a) We use the military as a row player and the attacker as a column player. The first value corresponds to the utility of the defender, while the second corresponds to the attacker's utility. The utility matrix in the normal-form game is then

		Attacker			
		$\mathbf{T}_1$	$\mathbf{T}_2$	$\mathbf{T}_3$	$\mathbf{T}_4$
Military	$\mathbf{T}_1$	$2,-1$	$0,7$	$0,1$	$0,5$
	$\mathbf{T}_2$	$0,3$	$2,-1$	$0,1$	$0,5$
	$\mathbf{T}_3$	$0,3$	$0,7$	$2,-1$	$0,5$
	$\mathbf{T}_4$	$0,3$	$0,7$	$0,1$	$2,-1$

(b) In the modified game, when the attacker attacks the defended base, the result is still the same. But when the base is nearby the protected base, the utility changes in the following way.

$$u_1(\mathbf{T}_i, \mathbf{T}_{i+1}) = \frac{1}{2}u_1(\mathbf{T}_{i+1}, \mathbf{T}_{i+1}) + \frac{1}{2}\cdot 2$$
$$u_2(\mathbf{T}_i, \mathbf{T}_{i+1}) = \frac{1}{2}u_2(\mathbf{T}_{i+1}, \mathbf{T}_{i+1}) + \frac{1}{2}\cdot (-1)$$

The utility matrix in the modified game is

		Attacker			
		$\mathbf{T}_1$	$\mathbf{T}_2$	$\mathbf{T}_3$	$\mathbf{T}_4$
Military	$\mathbf{T}_1$	$2,-1$	$1,3$	$0,1$	$1,2$
	$\mathbf{T}_2$	$1,1$	$2,-1$	$1,0$	$0,5$
	$\mathbf{T}_3$	$0,3$	$1,3$	$2,-1$	$1,2$
	$\mathbf{T}_4$	$1,1$	$0,7$	$1,0$	$2,-1$

(c) Neither game has any pure Nash equilibrium.

Solution 2.8

In order to remove a strategy during the iterated elimination, it has to be strictly dominated. Now, let us suppose that there exists a Nash equilibrium $\mathbf{s}^* \in \mathbf{S}$ in which player i plays strategy $s_i^* \in S_i$ that is strictly dominated by $s_i \in S_i$, that is,

$$u_i(s_i, \mathbf{s}_{-i}^*) > u_i(s_i^*, \mathbf{s}_{-i}^*) = u_i(\mathbf{s}^*).$$

This is in direct contradiction with the definition of a Nash equilibrium. Hence, the iterated elimination of strictly dominated strategies cannot remove any Nash equilibrium.

On the other hand, let us suppose that there exist a Nash equilibrium $\mathbf{s}^* \in \mathbf{S}$ in which player i plays strategy $s_i^* \in S_i$ that is only weakly dominated by $s_i \in S_i$ such that for some $\mathbf{s}'_{-i} \in \mathbf{S}_{-i}$ the following holds:

$$u_i(s_i, \mathbf{s}'_{-i}) > u_i(s_i^*, \mathbf{s}'_{-i}).$$

However, since $\mathbf{s}'_{-i} = \mathbf{s}^*_{-i}$ is not true in general, the above inequality does not necessarily violate the Nash equilibrium condition. This observation is instrumental in finding a counterexample with $\mathbf{s}'_{-i} \neq \mathbf{s}^*_{-i}$. For example:

		Player 2	
		d	e
Player 1	a	$0,1$	$1,0$
	b	$-1,0$	$1,2$
	c	$1,0$	$0,1$

Solution 2.9

(a) The only Pareto optimal strategies are $(\mathbf{D},\mathbf{R})$ and $(\mathbf{U},\mathbf{S})$.

(b) Pure Nash equilibria are $(\mathbf{D},\mathbf{R})$, $(\mathbf{U},\mathbf{V})$, $(\mathbf{C},\mathbf{V})$.

(c) Pure strategy **V** strictly dominates **S** (also **R** weakly dominates **S**).

		Player 2	
		V	**R**
Player 1	**U**	$1,3$	$-1,2$
	C	$1,0$	$0,-1$
	D	$1,2$	$3,3$

After removing **S**, both **U** and **C** are weakly dominated by **D**.

		Player 2	
		V	**R**
Player 1	**D**	$1,2$	$3,3$

Now, **R** strictly dominates **V**. This leaves us with only one strategy profile $(\mathbf{D},\mathbf{R})$, which is also a Nash equilibrium.

Solution 2.10

It is easy to verify that both strategy profiles (C,C) and (F,F) are equilibria in pure strategies. We will show that the game has a mixed strategy equilibrium, too.

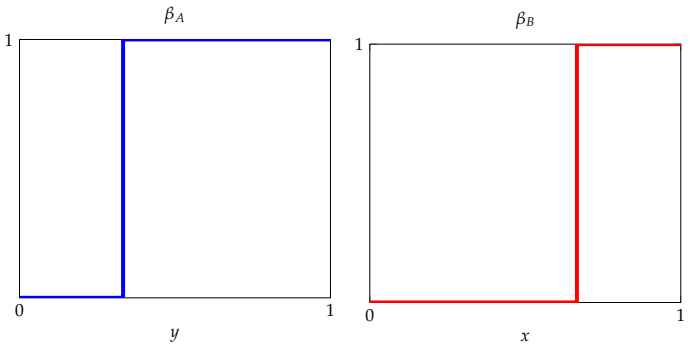
Let $x \in [0,1]$ and $y \in [0,1]$ be the probabilities of playing C for Alice and Bob, respectively. Since each player has only two pure strategies, the set of all mixed strategies can be viewed as the unit interval $[0,1]$. Thus, the expected utility of Alice is $U_A(x,y) = 2xy + (1-x)(1-y)$ and the expected utility of Bob is $U_B(x,y) = xy + 2(1-x)(1-y)$, for all $x,y \in [0,1]$.

We compute the best responses of both players. For Alice this is the mapping defined by

$$\beta_A(y) = \operatorname*{arg\,max}_{x\in[0,1]} U_A(x,y), \qquad y \in [0,1],$$

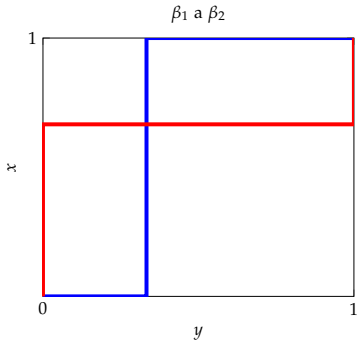
and analogously for Bob. We get

$$\beta_A(y) = \begin{cases} 0 & 0 \le y < \frac{1}{3}, \\ [0,1] & y = \frac{1}{3}, \\ 1 & \frac{1}{3} < y \le 1, \end{cases} \qquad \beta_B(x) = \begin{cases} 0 & 0 \le x < \frac{2}{3}, \\ [0,1] & x = \frac{2}{3}, \\ 1 & \frac{2}{3} < x \le 1. \end{cases}$$



We know that $(x^*, y^*) \in [0,1]^2$ would correspond to an equilibrium in mixed strategies if, and only if,

$$x^* \in \beta_A(y^*) \text{ and } y^* \in \beta_B(x^*). \tag{16}$$



The geometric interpretation of the condition (17) is that $(x^*, y^*) \in [0,1]^2$ is the point of the common intersection of the graphs of β_A a β_B. Therefore we obtain a mixed strategy equilibrium in which Alice plays the mixed strategy $(\frac{2}{3}, \frac{1}{3})$ and Bob uses the mixed strategy $(\frac{1}{3}, \frac{2}{3})$.

ALTERNATIVE SOLUTION: THE INDIFFERENCE PRINCIPLE. We can also find the mixed equilibrium directly, without drawing the best-response correspondences. Suppose that in an equilibrium *both* players play *both* strategies with positive probability, that is, $0 < x < 1$ and $0 < y < 1$. A strategy played with positive probability must be a best response: otherwise the player could move its probability to a better strategy and gain. Therefore both pure strategies of each player must give the *same* expected payoff against the opponent's mixed strategy.

Alice (C and F are in her support). Against Bob's mix y,

$$U_A(C,y) = 2y + 0 \cdot (1-y) = 2y, \qquad U_A(F,y) = 0 \cdot y + 1 \cdot (1-y) = 1-y.$$

Indifference $U_A(C,y) = U_A(F,y)$ gives $2y = 1-y$, so $y^* = \frac{1}{3}$.

Bob (C and F are in his support). Against Alice's mix x,

$$U_B(x,C) = 1 \cdot x + 0 \cdot (1-x) = x, \qquad U_B(x,F) = 0 \cdot x + 2 \cdot (1-x) = 2(1-x).$$

Indifference $U_B(x,C) = U_B(x,F)$ gives $x = 2(1-x)$, so $x^* = \frac{2}{3}$.

Check. Both x^* and y^* lie in $(0,1)$, so they are valid probabilities. Each player has only two strategies, so there is no strategy outside the support

that could pay more. Hence Alice plays $(\frac{2}{3}, \frac{1}{3})$ and Bob plays $(\frac{1}{3}, \frac{2}{3})$, the same equilibrium as above.

Note the cross-over: Alice's indifference equation contains only *Alice's* payoffs, yet its unknown is *Bob's* probability y. In a mixed equilibrium each player mixes so as to make the *opponent* indifferent, so each player's mix depends only on the opponent's payoffs.

The equilibrium payoffs are $U_A = 2y^* = \frac{2}{3}$ and $U_B = x^* = \frac{2}{3}$ (either strategy in the support gives the same value). Both players are worse off than in either pure equilibrium.

Can only one player mix? If Alice mixes while Bob plays a pure strategy, Alice must be indifferent against that pure strategy: against C this requires $2 = 0$, against F it requires $0 = 1$. Neither holds. Symmetrically, Bob mixing against a pure strategy of Alice would require $1 = 0$ or $0 = 2$. Hence the game has exactly three equilibria: (C, C), (F, F) and the mixed one.

Solution 2.11

(a) THE INDIFFERENCE PRINCIPLE. Suppose both drivers play both strategies with positive probability, $0 < p < 1$ and $0 < q < 1$. Then both S and D are best responses, so each driver must be indifferent between them.

Driver 1 (S and D in his support). Against Driver 2's mix q,

$$U_1(S, q) = 0 \cdot q + (-1)(1 - q) = q - 1,$$
$$U_1(D, q) = 1 \cdot q + (-10)(1 - q) = 11q - 10.$$

Indifference gives $q - 1 = 11q - 10$, so $q^* = \frac{9}{10}$.

Driver 2. The game is symmetric, so $U_2(S, p) = p - 1$ and $U_2(D, p) = 11p - 10$. Indifference gives $p^* = \frac{9}{10}$.

Check. Both values lie in $(0, 1)$, and there are no other strategies. Each driver swerves with probability $\frac{9}{10}$, and the equilibrium payoff is $U_1 = U_1(S, q^*) = q^* - 1 = -\frac{1}{10}$, the same for Driver 2.

Can only one driver mix? If Driver 1 mixes against a pure strategy of Driver 2, he must be indifferent against it: against S ($q = 1$) this means $0 = 1$, against D ($q = 0$) it means $-1 = -10$. Both are impossible, and the same holds for Driver 2. So the equilibria are (S, D), (D, S) and the mixed one.

(b) Again assume $0 < p < 1$ and $0 < q < 1$.

Driver 1 (motorcycle, crash -10). His payoffs are those of part (a):

$$U_1(S, q) = q - 1, \qquad U_1(D, q) = q - 10(1 - q) = 11q - 10,$$

so his indifference gives again $q^* = \frac{9}{10}$.

Driver 2 (armoured car, crash -5). Against Driver 1's mix p,

$$U_2(S, p) = 0 \cdot p + (-1)(1 - p) = p - 1,$$
$$U_2(D, p) = 1 \cdot p + (-5)(1 - p) = 6p - 5.$$

Indifference gives $p - 1 = 6p - 5$, so $p^* = \frac{4}{5}$.

Check. Both values lie in $(0, 1)$. Driver 1 drives straight with probability $1 - p^* = \frac{1}{5}$, Driver 2 with probability $1 - q^* = \frac{1}{10}$. The payoffs are $U_1 = q^* - 1 = -\frac{1}{10}$ and $U_2 = p^* - 1 = -\frac{1}{5}$.

Why does the motorcyclist go straight more often? His probability p^* comes from *Driver 2's* indifference equation, not from his own. Driver 2 fears the crash less, so Driver 1 has to go straight more often to make Driver 2 indifferent between swerving and going straight. Changing a driver's crash cost changes the *opponent's* equilibrium strategy, not his own.

(c) Assume $0 < p < 1$ and $0 < q < 1$. For Driver 1,

$$U_1(S, q) = q - 1, \qquad U_1(D, q) = q - c_1(1 - q).$$

Indifference gives $-1 = -c_1(1 - q)$, so $q^* = 1 - \frac{1}{c_1}$. In the same way, Driver 2's indifference $U_2(S, p) = p - 1 = p - c_2(1 - p) = U_2(D, p)$ gives $p^* = 1 - \frac{1}{c_2}$.

Check. Because $c_1, c_2 > 1$, both p^* and q^* lie in $(0, 1)$. (For $c_i \leq 1$ the mixed equilibrium would not exist.) Moreover, a driver mixing against a pure strategy would need $0 = 1$ or $-1 = -c_i$, which is impossible for $c_i > 1$.

Driver 1 drives straight with probability $1 - p^* = \frac{1}{c_2}$, and Driver 2 with probability $1 - q^* = \frac{1}{c_1}$. The payoffs are $U_1 = q^* - 1 = -\frac{1}{c_1}$ and $U_2 = p^* - 1 = -\frac{1}{c_2}$. For $c_1 = c_2 = 10$ this gives part (a), and for $c_1 = 10$, $c_2 = 5$ it gives part (b).

3 COMPUTING NASH EQUILIBRIA

Tutorial 3: Computing Nash equilibria **definitions**

Two players, mixed strategies p_1, p_2. $U_i(s_i, p_{-i})$ is player i's expected payoff from pure strategy s_i against the opponent's mix.

Best response.

$$\mathrm{BR}_i(p_{-i}) = \arg\max_{s_i} U_i(s_i, p_{-i}).$$

Nash equilibrium (mixed). p is a Nash equilibrium $\iff$ every strategy played with positive probability is a best response:

$$p_i(s_i) > 0 \implies s_i \in \mathrm{BR}_i(p_{-i}) \qquad \text{for both players.}$$

In 2×2 games: draw both best-response correspondences (steps) in the unit square; equilibria are the intersections.

Support enumeration. For every pair of supports (T_1, T_2) solve the linear feasibility problem (LP with constant objective) in p_1, p_2, e_1, e_2:

$$\begin{aligned}
&U_1(s_1, p_2) = e_1 && s_1 \in T_1 \quad \text{(indifference)}\\
&U_1(s_1, p_2) \le e_1 && s_1 \notin T_1 \quad \text{(no gain outside)}\\
&U_2(s_2, p_1) = e_2 && s_2 \in T_2\\
&U_2(s_2, p_1) \le e_2 && s_2 \notin T_2\\
&p_i \ge 0,\ p_i = 0 \text{ outside } T_i,\ \textstyle\sum p_i = 1.
\end{aligned}$$

Feasible $\Rightarrow$ Nash equilibrium. Infeasible $\Rightarrow$ next pair. A guess fails if (i) the equalities have no solution, (ii) some probability is negative, or (iii) a strategy outside the support pays more.

Quantal response (logit). With precision $\lambda \ge 0$,

$$\mathrm{QR}_i^{\lambda}(p_{-i})(s_i) = \frac{e^{\lambda U_i(s_i, p_{-i})}}{\sum_{t_i \in S_i} e^{\lambda U_i(t_i, p_{-i})}};$$

$$\text{For two strategies: } s, t \in S_i : P(s) = \frac{1}{1 + e^{-\lambda d}},\ d = U_i(s) - U_i(t).$$

$\lambda = 0$: uniform; $\lambda \to \infty$: best response.

Quantal response equilibrium (QRE). p is a QRE if $p = \mathrm{QR}^{\lambda}(p_{-i})$: each player plays the quantal response to the other's actual mix. As $\lambda \to \infty$, QRE approach Nash equilibria.

Nash vs. QRE: in a mixed Nash equilibrium a player mixes to make the *opponent* indifferent; in a QRE a player mixes according to *his own* payoff differences.

Exercise 3.12

Battle of the Sexes. Alice (rows) and Bob (columns) choose Concert (C) or Football (F).

	C	F
C	$(2,1)$	$(0,0)$
F	$(0,0)$	$(1,2)$

Let $x = P(\text{Alice plays } C)$ and $y = P(\text{Bob plays } C)$.

(a) Write Alice's expected payoff as a function of x and y. Draw both best-response correspondences in the unit square and read off all Nash equilibria.

(b) Find all Nash equilibria again by support enumeration: go through all 9 pairs of supports. Write the pair $(\{C,F\},\{C,F\})$ as a linear feasibility problem.

Exercise 3.13

Chicken. Two drivers head towards each other and choose Swerve (S) or Straight (D).

	S	D
S	$(0,0)$	$(-1,1)$
D	$(1,-1)$	$(-10,-10)$

Let $p = P(\text{driver 1 swerves})$ and $q = P(\text{driver 2 swerves})$.

(a) Write U_1 as a function of p and q. Draw both best-response correspondences and read off all Nash equilibria.

(b) In Battle of the Sexes, Alice's best response steps *up* as y grows; in Chicken, driver 1's best response steps *down* as q grows. Why?

(⋆c) Replace the crash payoff -10 by $-c$ with $c > 1$. In the mixed equilibrium, how often does each driver go straight, and what does each get?

Exercise 3.14

Patrol chase. The military defends one of four bases $T_1,\dots,T_4$; at the same time the attacker attacks one base. If the attacked base is defended, the military gets 2 and the attacker -1. Otherwise the military gets 0 and the attacker gets the value v_j of the base, where

$$(v_1,v_2,v_3,v_4) = (3,7,1,5).$$

Let d_j be the probability that the military defends T_j and a_j the probability that the attacker attacks T_j.

(a) Show that $U_M(T_j) = 2a_j$ and $U_A(T_j) = v_j - d_j(v_j+1)$.

(b) Guess that both players use the same support T. Show that the military's indifference forces the attacker to attack uniformly on T.

(c) Show that every base in T is worth at least as much as every base outside T. Which candidate sets T remain?

(d) Test the candidates and find the Nash equilibrium. Which base is never defended, and which is defended most?

Hint for (c): inside T the attacker earns $e_A = v_j - d_j(v_j + 1) \le v_j$; outside T he would earn v_j and this must not beat e_A.

Exercise 3.15

A degenerate game.

	L	R
T	$(1,1)$	$(0,0)$
B	$(1,0)$	$(2,1)$

(a) Find all Nash equilibria.

(b) Which of them would support enumeration miss if it tested only supports of equal size? Explain by counting equations.

Exercise 3.16

Quantal response in asymmetric Matching Pennies. Alice (rows) wants to match Bob (columns), Bob wants to mismatch. Alice's payoff at (T, L) is 9 instead of 1.

	L	R
T	$(9,0)$	$(0,1)$
B	$(0,1)$	$(1,0)$

Let $x = P(\text{Alice plays } T)$ and $y = P(\text{Bob plays } L)$. The logit quantal response of player i with precision $\lambda \ge 0$ plays s_i with probability

$$\mathrm{QR}_i^{\lambda}(p_{-i})(s_i) = \frac{\exp\left(\lambda\, U_i(s_i, p_{-i})\right)}{\sum_{t_i} \exp\left(\lambda\, U_i(t_i, p_{-i})\right)},$$

and a profile p is a quantal response equilibrium (QRE) if $p = \mathrm{QR}^{\lambda}(p)$.

(a) Find the Nash equilibrium. How does Alice's payoff 9 influence her own equilibrium strategy?

(b) Show that Alice's quantal response to y is

$$P(T) = \frac{1}{1 + e^{-\lambda d}}, \qquad d = U_A(T) - U_A(B) = 10y - 1.$$

For $\lambda = 1$, compute it for $y = 0,\ 0.1,\ 0.3$ and compare with Alice's best response.

(c) Let $\lambda = 1$. Which of the following profiles is a QRE?

A: $(x, y) = (0.8,\ 0.3)$, B: $(x, y) = (0.5,\ 0.1)$, C: $(x, y) = (0.894,\ 0.313)$.

(d) Compare the Nash equilibrium with the QRE. Why is the Nash equilibrium not a QRE?

SOLUTIONS

Solution 3.12

(a) It is easy to verify that both strategy profiles (C, C) and (F, F) are equilibria in pure strategies. We will show that the game has a mixed strategy equilibrium, too.

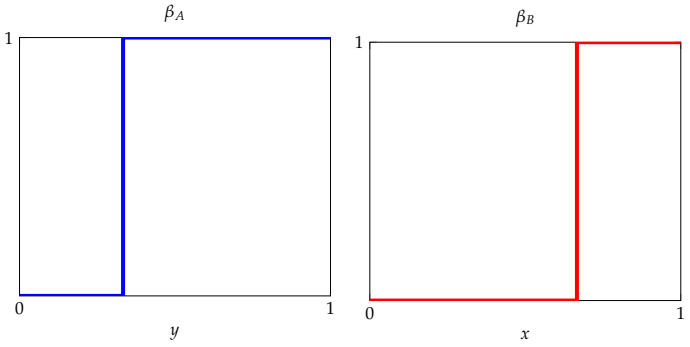
Let $x \in [0,1]$ and $y \in [0,1]$ be the probabilities of playing C for Alice and Bob, respectively. Since each player has only two pure strategies, the set of all mixed strategies can be viewed as the unit interval $[0,1]$. Thus, the expected utility of Alice is $U_A(x,y) = 2xy + (1-x)(1-y)$ and the expected utility of Bob is $U_B(x,y) = xy + 2(1-x)(1-y)$, for all $x, y \in [0,1]$.

We compute the best responses of both players. For Alice this is the mapping defined by

$$\beta_A(y) = \arg\max_{x \in [0,1]} U_A(x,y), \qquad y \in [0,1],$$

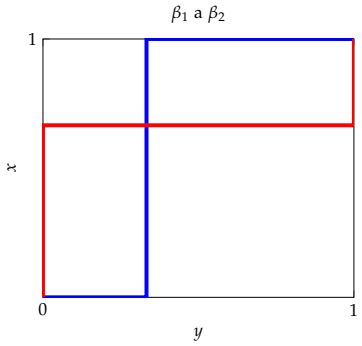
and analogously for Bob. We get

$$\beta_A(y) = \begin{cases} 0 & 0 \le y < \frac{1}{3}, \\ [0,1] & y = \frac{1}{3}, \\ 1 & \frac{1}{3} < y \le 1, \end{cases} \qquad \beta_B(x) = \begin{cases} 0 & 0 \le x < \frac{2}{3}, \\ [0,1] & x = \frac{2}{3}, \\ 1 & \frac{2}{3} < x \le 1. \end{cases}$$



We know that $(x^*, y^*) \in [0,1]^2$ would correspond to an equilibrium in mixed strategies if, and only if,

$$x^* \in \beta_A(y^*) \text{ and } y^* \in \beta_B(x^*). \tag{17}$$



The geometric interpretation of the condition (17) is that $(x^*, y^*) \in [0,1]^2$ is the point of the common intersection of the graphs of β_A a β_B. Therefore we

obtain a mixed strategy equilibrium in which Alice plays the mixed strategy $(\frac{2}{3}, \frac{1}{3})$ and Bob uses the mixed strategy $(\frac{1}{3}, \frac{2}{3})$.

(b) The nine pairs fall into four groups.

- $(\{C\}, \{C\})$ and $(\{F\}, \{F\})$: no one gains by deviating ($2 \geq 0$, $1 \geq 0$; $1 \geq 0$, $2 \geq 0$). Equilibria.
- $(\{C\}, \{F\})$ and $(\{F\}, \{C\})$: Alice gets 0 but the other row gives 1 or 2. The outside inequality fails.
- One pure, one mixed, e.g. $(\{C\}, \{C, F\})$: Bob must be indifferent against C, i.e. $1 = 0$. The equalities contradict each other. The same happens for the other three such pairs ($0 = 2$, $2 = 0$, $0 = 1$).
- $(\{C, F\}, \{C, F\})$: find x, y, e_1, e_2 with

  $$2y - e_1 = 0, \quad (1-y) - e_1 = 0, \quad x - e_2 = 0, \quad 2(1-x) - e_2 = 0, \quad 0 \leq x, y \leq 1.$$

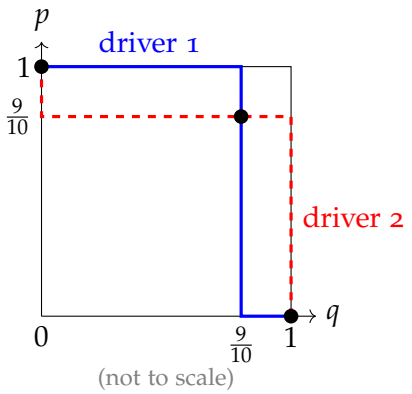
  This is an LP with a constant objective. Solution $y = \frac{1}{3}$, $x = \frac{2}{3}$, $e_1 = e_2 = \frac{2}{3}$. No outside inequalities, since both supports are full.

Solution 3.13

(a) $U_1(S) = q \cdot 0 + (1-q)(-1) = q - 1$ and $U_1(D) = q - 10(1-q) = 11q - 10$, so

$$U_1(p, q) = p(q-1) + (1-p)(11q - 10) = p(9 - 10q) + 11q - 10.$$

Driver 1 swerves for $q < \frac{9}{10}$, goes straight for $q > \frac{9}{10}$, anything at $q = \frac{9}{10}$. Driver 2 is symmetric.



Three equilibria: (S, D), (D, S), and both swerve with probability $\frac{9}{10}$, each getting $q - 1 = -\frac{1}{10}$.

(b) In BoS Alice wants to *match* Bob: the more Bob goes to the concert, the better C is for her (coefficient $3y - 1$ grows in y). In Chicken a driver wants to *do the opposite*: the more the other swerves, the better straight is (coefficient $9 - 10q$ falls in q).

($\star$c) $U_1(S) = q - 1$, $U_1(D) = q - c(1-q)$. Equal iff $q = 1 - \frac{1}{c}$. So each driver goes straight with probability $\frac{1}{c}$ and gets $-\frac{1}{c}$. A more dangerous crash makes each driver go straight less often, and the expected loss shrinks.

Solution 3.14

(a) Defending T_j pays 2 exactly when T_j is attacked: $U_M(T_j) = 2a_j$. Attacking T_j pays -1 with probability d_j and v_j otherwise: $U_A(T_j) = -d_j + (1-d_j)v_j = v_j - d_j(v_j+1)$.

(b) The military is indifferent on T: $2a_j = e_M$ for all $j \in T$. So all a_j on T are equal: $a_j = \frac{1}{|T|}$ and $e_M = \frac{2}{|T|}$. The attacker's mix is fixed by the *military's* payoffs; the values v_j do not enter.

(c) Inside T: $e_A = v_j - d_j(v_j+1)$. Outside T: $d_j = 0$, so attacking pays v_j, and we need $v_j \le e_A$. Hence every value of the base inside the support is $\ge e_A$ and every value of base outside the support $\le e_A$: T consists of the most valuable bases. Ordered by value $T_2(7)$, $T_4(5)$, $T_1(3)$, $T_3(1)$, the candidates are

$$\{T_2\}, \quad \{T_2, T_4\}, \quad \{T_1, T_2, T_4\}, \quad \{T_1, T_2, T_3, T_4\}.$$

(d) The attacker's indifference gives $d_j = \dfrac{v_j - e_A}{v_j + 1}$ on T, and $\sum_{j\in T} d_j = 1$ fixes e_A.

- $\{T_2\}$: $d_2 = 1$, $e_A = 7 - 8 = -1$. Attacking T_4 pays $5 > -1$. Rejected.
- $\{T_2, T_4\}$: $a_2 = a_4 = \frac{1}{2}$. From $\frac{7-e_A}{8} + \frac{5-e_A}{6} = 1$ we get $e_A = \frac{17}{7}$ and $d = (\frac{4}{7}, \frac{3}{7})$. Attacking T_1 pays $3 > \frac{17}{7}$. Rejected as NE.
- $\{T_1, T_2, T_4\}$: $a_1 = a_2 = a_4 = \frac{1}{3}$. From $\frac{3-e_A}{4} + \frac{7-e_A}{8} + \frac{5-e_A}{6} = 1$ which gets us $e_A = \frac{35}{13}$, so

  $$(d_1, d_2, d_4) = \left(\tfrac{1}{13}, \tfrac{7}{13}, \tfrac{5}{13}\right).$$

  Outside: attacking T_3 pays $1 \le \frac{35}{13}$; defending T_3 pays $0 \le \frac{2}{3} = e_M$. Since the probabilities ar valid we have the **Equilibrium.**
- All four: $e_A = \frac{47}{25}$ and $d_3 = \frac{1-47/25}{2} = -\frac{11}{25} < 0$. Rejected.

Nash equilibrium: the military defends T_1, T_2, T_4 with probabilities $\frac{1}{13}, \frac{7}{13}, \frac{5}{13}$; the attacker attacks T_1, T_2, T_4 with probability $\frac{1}{3}$ each. T_3 is never defended (and never attacked); T_2 is defended most. Payoffs: military $\frac{2}{3}$, attacker $\frac{35}{13}$.

Every step was an indifference condition, except the rejections: $\{T_2\}$ and $\{T_2, T_4\}$ fail the outside inequality, the full set fails non-negativity.

Solution 3.15

(a) Let $x = P(T)$, $y = P(L)$. Alice gets $U_A(T) = y$ and $U_A(B) = y + 2(1-y) = 2 - y$, so B is strictly better unless $y = 1$, where she is indifferent. Bob gets $U_B(L) = x$ and $U_B(R) = 1 - x$.

- If $y < 1$, Alice plays B, so Bob plays R: the equilibrium (B, R).
- If $y = 1$, any x is a best response for Alice, and L is a best response for Bob iff $x \ge 1 - x$, i.e. $x \ge \frac{1}{2}$.

Nash equilibria: (B, R) and the whole segment $\{(x\,T + (1-x)\,B,\ L) : \frac{1}{2} \le x \le 1\}$, infinitely many.

(b) The segment uses supports $\{T, B\}$ and $\{L\}$ (for $x < 1$), of different sizes. Alice's mix has one free number; it would be pinned by Bob's indifference, but Bob uses a single strategy, so there is no equation. Only the inequality $U_B(R) \leq U_B(L)$ bounds x, which gives an interval. Equal-size enumeration finds only (T, L) and (B, R).

The cause is a tie: Bob's pure strategy L (one strategy) has two best responses for Alice, T and B. A game where some mixed strategy with k strategies in its support has more than k pure best responses is called *degenerate*.

Solution 3.16

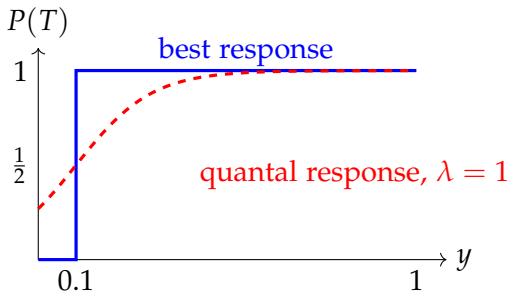
(a) Alice is indifferent iff $U_A(T) = 9y$ equals $U_A(B) = 1 - y$, i.e. $y = \frac{1}{10}$. Bob is indifferent iff $U_B(L) = 1 - x$ equals $U_B(R) = x$, i.e. $x = \frac{1}{2}$. The unique Nash equilibrium is $x = \frac{1}{2}$, $y = \frac{1}{10}$. Alice's payoff 9 changes only Bob's strategy: Alice still plays T with probability $\frac{1}{2}$, exactly as in ordinary Matching Pennies, because her mix is pinned by Bob's indifference.

(b) Dividing the whole fraction by $e^{\lambda U_A(T)}$,

$$p(T) = \frac{e^{\lambda \cdot 9y}}{e^{\lambda \cdot 9y} + e^{\lambda(1-y)}} = \frac{1}{1 + e^{-\lambda(10y-1)}}.$$

Only the payoff difference $d_A = 10y - 1$ matters. For $\lambda = 1$:

y	d_A	best response $P(T)$	quantal response $P(T)$
0	-1	0	0.27
0.1	0	any $x \in [0, 1]$	0.5
0.3	2	1	0.88



At $y = 0.1$ the best response is any mix, the quantal response is exactly $\frac{1}{2}$. For $\lambda = 0$ the curve is flat at $\frac{1}{2}$; for $\lambda \to \infty$ it becomes the step.

(c) Check both players: Alice's quantal response to y must return x (with $d_A = 10y - 1$), and Bob's quantal response to x must return y (with $d_B = U_B(L) - U_B(R) = 1 - 2x$).

	Alice: $d_A \to P(T)$	Bob: $d_B \to P(L)$	
A	$2 \to 0.881 \neq 0.8$	$-0.6 \to 0.354 \neq 0.3$	not a QRE
B	$0 \to 0.5 = 0.5$ ✓	$0 \to 0.5 \neq 0.1$	not a QRE
C	$2.13 \to 0.894$ ✓	$-0.788 \to 0.313$ ✓	QRE

(d) The two predictions differ considerably:

	Nash	QRE ($\lambda = 1$)
$x = P(T)$	0.5	0.894
$y = P(L)$	0.1	0.313

In the QRE Alice's own payoff 9 matters: she plays T much more often. Bob anticipates this and mostly plays R, but being noisy he still plays L with probability 0.313. As λ grows, the QRE approaches the Nash equilibrium: $(0.702, 0.117)$ for $\lambda = 5$ and $(0.522, 0.100)$ for $\lambda = 50$.

Tutorial 4: Zero-sum games

definitions

Matrix (two-player zero-sum) game. $u_1(s_1, s_2) + u_2(s_1, s_2) = 0$ for all (s_1, s_2). Write $u = u_1$ and $A_{ij} = u(s_1^i, s_2^j)$: player 1 (rows) *maximizes*, player 2 (columns) *minimizes*. Expected payoff $U(p_1, p_2) = p_1^\top A p_2$.

Pure saddle point. (s_1^*, s_2^*) with $u(s_1, s_2^*) \le u(s_1^*, s_2^*) \le u(s_1^*, s_2)$ for all s_1, s_2: the entry is the *minimum of its row* and the *maximum of its column*. It exists iff $\max_{s_1} \min_{s_2} u(s_1, s_2) = \min_{s_2} \max_{s_1} u(s_1, s_2)$ (pure maximin = pure minimax).

Maximin and minimax values.

$$\underline{v} = \max_{p_1 \in \Delta_1} \min_{p_2 \in \Delta_2} U(p_1, p_2) = \max_{p_1 \in \Delta_1} \min_{s_2 \in S_2} U(p_1, s_2),$$
$$\overline{v} = \min_{p_2 \in \Delta_2} \max_{p_1 \in \Delta_1} U(p_1, p_2) = \min_{p_2 \in \Delta_2} \max_{s_1 \in S_1} U(s_1, p_2).$$

The inner optimization may use *pure* strategies only: for fixed p_1, $U(p_1, p_2)$ is linear in p_2, so its minimum over Δ_2 is attained at a pure strategy s_2 (and similarly for the maximum over Δ_1). An optimal p_1^* is a *maximin strategy* (the best guarantee of player 1), an optimal p_2^* a *minimax strategy*. Always $\underline{v} \le \overline{v}$.

Minimax theorem (von Neumann). Every matrix game has $\underline{v} = \overline{v} =: v$, the *value of the game*. Moreover: (p_1^*, p_2^*) is a Nash equilibrium $\iff$ p_1^* is maximin and p_2^* is minimax. All equilibria give the payoff v, and optimal strategies are *interchangeable*: any maximin strategy with any minimax strategy is an equilibrium.

The two linear programs.

maximize	v_1	minimize	v_2
subject to	$A^\top p_1 \ge v_1 \mathbf{1}_n$	subject to	$A\, p_2 \le v_2 \mathbf{1}_m$
	$\mathbf{1}_m^\top p_1 = 1,\ p_1 \ge \mathbf{0}$		$\mathbf{1}_n^\top p_2 = 1,\ p_2 \ge \mathbf{0}$

One constraint per pure strategy of the opponent. The two LPs are dual; their optimal values are equal, $v_1^* = v_2^* = v$. Feasible solutions with equal objective values are optimal.

Double Oracle. Start with small sets $T_1 \subseteq S_1$, $T_2 \subseteq S_2$ and repeat:

1. *Restricted game:* solve the game with strategies T_1, T_2 only (the two LPs); equilibrium (q_1^*, q_2^*).
2. *Oracles:* best responses in the *full* game, $s_1 \in \arg\max_{t_1 \in S_1} U(t_1, q_2^*)$ and $s_2 \in \arg\min_{t_2 \in S_2} U(q_1^*, t_2)$.
3. *Stop* if $s_1 \in T_1$ *and* $s_2 \in T_2$: then (q_1^*, q_2^*) is an equilibrium of the full game.
4. *Expand:* $T_1 \leftarrow T_1 \cup \{s_1\}$, $T_2 \leftarrow T_2 \cup \{s_2\}$.

4 ZERO–SUM GAMES

Exercise 4.17

Colonel Blotto and Colonel Baloney. Each colonel has 2 units and secretly distributes them over three bases worth 1, 2 and 3 points. Whoever has more units on a base takes its points from the opponent; with equal units nobody pays. A strategy is written (x_1, x_2, x_3), the units on bases $1, 2, 3$.

(a) List the strategies and write Blotto's payoff matrix of this zero-sum game.

(b) Compute the pure maximin and minimax values. Is there a saddle point? What is the value of the game?

Exercise 4.18

Alice and Bob play one round of a zero-sum game captured by the payoff matrix of Alice:

	e	f	g
a	6	0	-1
b	5	4	9
c	9	-3	-1
d	-1	1	-1

(a) Compute the pure maximin and pure minimax values. Does the game have a pure saddle point?

(b) Alice reveals publicly that she will be using strategy b. After making her choice public she must stick to it. Can Bob take advantage of knowing the strategic choice of Alice compared to the standard situation when Alice's strategic choice wouldn't be known a priori?

Exercise 4.19

The two-player zero-sum game with the payoff matrix for the first player

Player 1 \ Player 2	0	1
0	1	-1
1	-1	1

is called *Matching Pennies*. In this game, each player chooses one bit (or a side of the coin), 0 or 1, in the following way: each player inserts into an envelope a slip of paper on which his choice is written. The envelopes are sealed and submitted to a referee. If both players have selected the same bit, Player 2 pays one dollar to Player 1. If they have selected opposite bits, Player 1 pays one dollar to Player 2. Find an equilibrium of this game.

Exercise 4.20

Alice and Bob play Rock–Paper–Scissors, but Bob's fingers hurt preventing him from signalling "Scissors". Model this scenario as a zero-sum game and find its equilibrium.

★ **Exercise 4.21**
Consider a two-person zero-sum game with the payoff matrix

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \qquad a, b, c, d \in \mathbb{R}.$$

Find equilibrium strategies of the row and column player.

★ **Exercise 4.22**
Show that the following two-player zero-sum game doesn't have an equilibrium in pure strategies. The strategy space of each player is the set $X = \{0, \frac{1}{10}, \ldots, \frac{9}{10}, 1\}$ and the payoff function of Player 1 is

$$u(s_1, s_2) = \frac{1}{1 + (s_1 - s_2)^2}, \qquad s_1, s_2 \in X.$$

Exercise 4.23
Double Oracle. Consider again the game of Colonel Blotto and Colonel Baloney, but pretend that its payoff matrix is far too large to write down: we can only solve small restricted games and ask an oracle for a best response.

(a) Run the Double Oracle algorithm from $T_1 = T_2 = \{(1, 1, 0)\}$. In every iteration give the restricted game and its equilibrium, the oracles' best responses, and the termination check.

(b) Run the algorithm again, now from $T_1 = \{(1, 1, 0)\}$ for Blotto and $T_2 = \{(0, 1, 1)\}$ for Baloney. If an oracle has several best responses, take $(0, 0, 2)$. Can the algorithm stop when only *one* of the oracles returns a strategy that is already in its set?

SOLUTIONS

Solution 4.17

(a) There are six strategies. Blotto's payoff (his points minus Baloney's):

	$(0,0,2)$	$(0,1,1)$	$(0,2,0)$	$(1,0,1)$	$(1,1,0)$	$(2,0,0)$	min
$(0,0,2)$	0	1	1	2	0	2	**0**
$(0,1,1)$	-1	0	1	1	2	4	-1
$(0,2,0)$	-1	-1	0	-2	1	1	-2
$(1,0,1)$	-2	-1	2	0	1	2	-2
$(1,1,0)$	0	-2	-1	-1	0	1	-2
$(2,0,0)$	-2	-4	-1	-2	-1	0	-4
max	**0**	1	2	2	2	4	

For example, $(1,1,0)$ against $(0,1,1)$: base 1 is won $(+1)$, base 2 is tied, base 3 is lost (-3), so -2.

(b) The pure maximin value is 0 (row $(0,0,2)$) and the pure minimax value is 0 (column $(0,0,2)$). So both colonels putting everything on the most valuable base is a saddle point and the value is 0. This can be seen even before computing: the game is symmetric, its matrix satisfies $A = -A^\top$, and every symmetric zero-sum game has value 0 (each player can copy the other's optimal strategy). Concentrating on base 3 is the *safe* strategy: it never loses. In a zero-sum game the safe strategy is an equilibrium strategy.

Solution 4.18

(a) Row minima are $-1, 4, -3, -1$, so the pure maximin value is 4 (row b). Column maxima are $9, 4, 9$, so the pure minimax value is 4 (column f). They are equal, so (b, f) is a saddle point: the entry 4 is the smallest in its row and the largest in its column. The value of the game is 4.

(b) No. Knowing that Alice plays b, Bob's best response is f, and Alice gets 4. Without the announcement Bob would play his minimax strategy f anyway, which holds Alice to at most 4 whatever she does. In a zero-sum game, announcing an optimal (maximin) strategy never helps the opponent: it guarantees the value against *every* reply, including the best one.

Solution 4.19

Matching Pennies is a two-player zero-sum game, so we can formulate the equilibrium problem as a linear programming problem, as for restricted Rock–Paper–Scissors. However, we will first take a different approach and find the solution in a more elementary way.

First, it can be easily checked that the game has no solution in pure strategies. Let $x \in [0,1]$ and $y \in [0,1]$ denote the probabilities of selecting zero bit for Player 1 and 2, respectively. The expected utility of Player 1 is then the function given by $U(x,y) = 4xy - 2x - 2y + 1$, for every $x, y \in [0,1]$. For any choice $x \in [0,1]$ of Player 1, Player 2 will select the most harmful strategy for Player 1. This implies that Player 1 gets in this worst case scenario

$$\ell(x) := \min_{y \in [0,1]} U(x,y) = \begin{cases} 2x - 1, & 0 \le x < \frac{1}{2}, \\ 0, & x = \frac{1}{2}, \\ 1 - 2x, & \frac{1}{2} < x \le 1, \end{cases} \qquad x \in [0,1].$$

Player 1 can secure the maxmin value

$$\max_{x\in[0,1]} \ell(x), \tag{18}$$

which is equal to the minmax value of Player 2 by von Neumann's theorem. The equilibrium strategy of Player 1 is then any solution to the optimization problem (18). The only such solution is $x^* = \frac{1}{2}$. Repeating this analysis for Player 2 and the minmax value, we arrive at the same solution $y^* = \frac{1}{2}$ for Player 2. In conclusion, the only equilibrium strategy for each player is to randomize uniformly between the two choices.

The same solution can be obtained from the linear programs of both players. With $x = (x_1, x_2)$ and $y = (y_1, y_2)$ the probabilities of bits $0, 1$:

$$\begin{array}{ll} \text{maximize} & x_0 \\ \text{subject to} & x_1 - x_2 \ \geq x_0 \quad \text{(column 0)} \\ & -x_1 + x_2 \ \geq x_0 \quad \text{(column 1)} \\ & x_1 + x_2 = 1, \ \ x_1, x_2 \geq 0, \end{array} \qquad \begin{array}{ll} \text{minimize} & y_0 \\ \text{subject to} & y_1 - y_2 \ \leq y_0 \quad \text{(row 0)} \\ & -y_1 + y_2 \ \leq y_0 \quad \text{(row 1)} \\ & y_1 + y_2 = 1, \ \ y_1, y_2 \geq 0. \end{array}$$

The strategies $x = y = (\frac{1}{2}, \frac{1}{2})$ with $x_0 = y_0 = 0$ are feasible, and the objective values are equal, so both are optimal and the value of the game is 0.

Solution 4.20

This is a matrix game which can be described by the payoff matrix for Alice:

$$\mathbf{A} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \\ -1 & 1 \end{bmatrix}.$$

Since this is a two-person zero-sum game, its solution can be recovered by two dual linear programming problems. For Alice we solve the problem with variables x_0 and $\mathbf{x} = (x_1, x_2, x_3)$:

$$\begin{array}{ll} \text{Maximize} & x_0 \\ \text{subject to} & \mathbf{A}^\top \mathbf{x} - \mathbf{1} x_0 \geq \mathbf{0}, \\ & \sum_{i=1}^3 x_i = 1, \\ & \mathbf{x} \geq \mathbf{0}. \end{array}$$

The problem for Bob has variables y_0 and $\mathbf{y} = (y_1, y_2)$:

$$\begin{array}{ll} \text{Minimize} & y_0 \\ \text{subject to} & \mathbf{A}\mathbf{y} - \mathbf{1} y_0 \leq \mathbf{0}, \\ & \sum_{i=1}^2 y_i = 1, \\ & \mathbf{y} \geq \mathbf{0}. \end{array}$$

Equivalently, we can write:

$$\begin{array}{ll} \text{Maximize} & x_0 \\ \text{subject to} & x_2 - x_3 - x_0 \geq 0, \\ & -x_1 + x_3 - x_0 \geq 0, \\ & x_1 + x_2 + x_3 = 1, \\ & x_1, x_2, x_3 \geq 0, \end{array} \qquad \text{and} \qquad \begin{array}{ll} \text{Minimize} & y_0 \\ \text{subject to} & -y_2 - y_0 \leq 0, \\ & y_1 - y_0 \leq 0, \\ & -y_1 + y_2 - y_0 \leq 0, \\ & y_1 + y_2 = 1, \\ & y_1, y_2 \geq 0. \end{array}$$

The solutions of those two problems are

$$\mathbf{x}^* = (0, \tfrac{2}{3}, \tfrac{1}{3}) \quad \text{and} \quad \mathbf{y}^* = (\tfrac{1}{3}, \tfrac{2}{3}).$$

The value of game is equal to the common value in the optima, $x_0^* = y_0^* = \frac{1}{3}$.

Graphical solution. Bob has only two strategies, so after substituting $y_2 = 1 - y_1$ his program has just two variables, $y = y_1$ (the probability of Rock) and y_0. His three constraints become

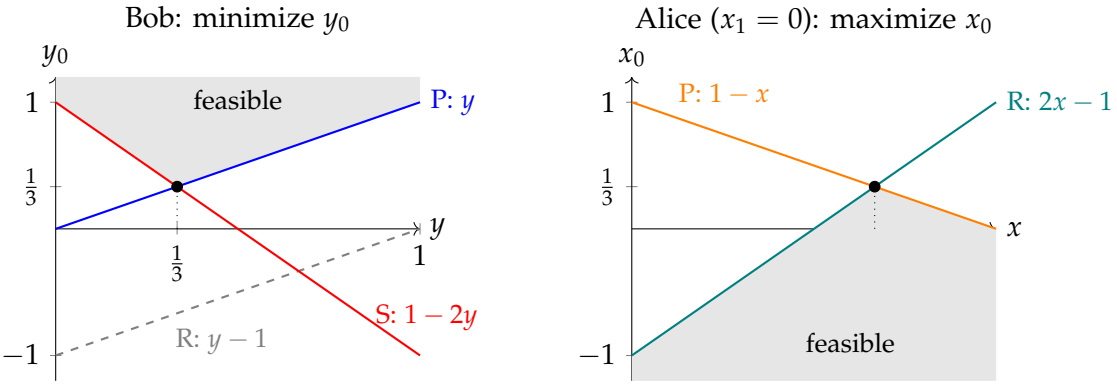
$$y_0 \geq y - 1 \quad \text{(Alice plays Rock)}, \qquad y_0 \geq y \quad \text{(Paper)}, \qquad y_0 \geq 1 - 2y \quad \text{(Scissors)}.$$

Each constraint is a line: Alice's expected payoff from one of her pure strategies against Bob's mixed strategy. The feasible region lies above all three lines, i.e. above their upper envelope (left picture). Minimizing y_0 means finding the lowest point of this region, which is $y = \frac{1}{3}$, $y_0 = \frac{1}{3}$. The Rock line lies below the Paper line everywhere, so this constraint is never active.

Alice has three strategies, so her program has three probabilities. However, Rock is strictly dominated by Paper ($0 < 1$ and $-1 < 0$), so $x_1 = 0$ in every equilibrium strategy. Substituting $x_1 = 0$ and $x = x_2$ (the probability of Paper), $x_3 = 1 - x$, her two constraints become

$$x_0 \leq 2x - 1 \quad \text{(Bob plays Rock)}, \qquad x_0 \leq 1 - x \quad \text{(Bob plays Paper)}.$$

The feasible region lies below both lines, i.e. below their lower envelope (right picture), and maximizing x_0 gives its highest point $x = \frac{2}{3}$, $x_0 = \frac{1}{3}$. Note that x_0 and y_0 are free variables: the value of a game may be negative.



★ Solution 4.21

Let x and y be the probabilities of the first row and the first column. The row player's expected payoff is

$$U(x, y) = (a + d - b - c)\, xy + (b - d)\, x + (c - d)\, y + d.$$

Case 1: A has a pure saddle point (an entry that is the minimum of its row and the maximum of its column). Then that pure profile is an equilibrium. This always happens if $a + d - b - c = 0$: then U is a sum of a function of x and a function of y, so each player has a dominant strategy. For example, $A = \begin{pmatrix} 3 & 2 \\ 2 & 1 \end{pmatrix}$ has the saddle point (first row, second column) with value 2.

Case 2: no pure saddle point. Then the equilibrium is fully mixed: if one player played a pure strategy, the other's pure best response would form a saddle point. So the equilibrium (x^*, y^*) lies in the interior $(0,1)^2$. By von Neumann's minimax theorem an equilibrium exists, and it is a saddle point of the function U on $[0,1]^2$:

$$U(x, y^*) \le U(x^*, y^*) \le U(x^*, y) \qquad \text{for all } x, y \in [0,1].$$

An interior saddle point satisfies the condition

$$\frac{\partial U}{\partial x}(x^*, y^*) = \frac{\partial U}{\partial y}(x^*, y^*) = 0,$$

which reads as

$$(a + d - b - c)\, y^* + b - d = (a + d - b - c)\, x^* + c - d = 0.$$

If $a + d - b - c = 0$, these equations have no solution unless $b = d$ and $c = d$; this is Case 1. So $a + d - b - c \neq 0$, and the only solution is

$$\mathbf{x}^* = (x^*, 1 - x^*)^\top = \frac{1}{a + d - b - c}\,(d - c,\ a - b)^\top, \qquad \mathbf{y}^* = (y^*, 1 - y^*)^\top = \frac{1}{a + d - b - c}\,(d - b,\ a -$$

Since the Hessian of U

$$\begin{pmatrix} 0 & a + d - b - c \\ a + d - b - c & 0 \end{pmatrix}$$

is indefinite, (x^*, y^*) is indeed a saddle point of U (neither a maximum nor a minimum). The value of the game is

$$v = U(x^*, y^*) = \frac{ad - bc}{a + d - b - c}.$$

(The same formulas follow from the indifference conditions: y^* makes the row player indifferent between his rows, and x^* makes the column player indifferent between his columns.)

Examples. Matching Pennies ($a = d = 1$, $b = c = -1$): $x^* = y^* = \frac{2}{4} = \frac{1}{2}$ and $v = \frac{1-1}{4} = 0$. For $A = \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix}$: $x^* = y^* = \frac{2}{5}$ and $v = \frac{2-1}{5} = \frac{1}{5}$.

Warning. The formulas are valid only in Case 2. For $A = \begin{pmatrix} 4 & 1 \\ 2 & 0 \end{pmatrix}$, which has a saddle point (first row, second column, value 1), they would give $x^* = -2$, which is not a probability. Always check for a saddle point first.

★ **Solution 4.22**

It suffices to show that the pure maximin value is strictly smaller than the pure minimax value. Player 1 wants to be close to Player 2, Player 2 wants to be far away. For fixed s_1, the worst s_2 is the endpoint farthest from s_1:

$$\min_{s_2} u(s_1, s_2) = \begin{cases} \dfrac{1}{1 + (1 - s_1)^2}, & s_1 \le \frac{1}{2}, \\ \dfrac{1}{1 + s_1^2}, & s_1 > \frac{1}{2}, \end{cases}$$

which is largest at $s_1 = \frac{1}{2}$, so the pure maximin value is $\frac{1}{1+1/4} = \frac{4}{5}$. For fixed s_2, Player 1 can copy it and get $u = 1$, so the pure minimax value is 1. Since $\frac{4}{5} < 1$, there is no saddle point and no pure equilibrium. By the minimax theorem, a mixed equilibrium exists.

Solution 4.23

(a) In this small game an oracle reads one column or one row of the payoff matrix of the Blotto and Baloney game. Each oracle table lists Blotto's payoff for every allocation of the player who is answering, against the opponent's current strategy; the best answer is in bold.

Iteration 1.

- *We have* $T_1 = \{(1,1,0)\}$ and $T_2 = \{(1,1,0)\}$.
- *Restricted game.* 1×1: both must play $(1,1,0)$. Value 0.
- *Blotto's oracle* (largest payoff = best for Blotto). Best reply to Baloney's $(1,1,0)$:

Blotto	$(0,0,2)$	$(0,1,1)$	$(0,2,0)$	$(1,0,1)$	$(1,1,0)$	$(2,0,0)$
vs $(1,1,0)$	0	**2**	1	1	0	-1

  Best: $(0,1,1)$ with 2 (base 1 lost, base 2 tied, base 3 won: $-1+0+3$). New.

- *Baloney's oracle* (smallest payoff = best for Baloney). Best reply to Blotto's $(1,1,0)$:

Baloney	$(0,0,2)$	$(0,1,1)$	$(0,2,0)$	$(1,0,1)$	$(1,1,0)$	$(2,0,0)$
vs $(1,1,0)$	0	$-$**2**	-1	-1	0	1

  Best: $(0,1,1)$ with -2. New.

- *Decision.* Both best responses are new: add $(0,1,1)$ to both sets.

Iteration 2.

- *We have* $T_1 = \{(1,1,0),(0,1,1)\}$ and $T_2 = \{(1,1,0),(0,1,1)\}$.
- *Restricted game.* $\begin{pmatrix} 0 & -2 \\ 2 & 0 \end{pmatrix}$ (rows and columns $(1,1,0),(0,1,1)$). $(0,1,1)$ is better for both players, so both play $(0,1,1)$. Value 0.
- *Blotto's oracle* (largest payoff = best for Blotto). Best reply to $(0,1,1)$:

Blotto	$(0,0,2)$	$(0,1,1)$	$(0,2,0)$	$(1,0,1)$	$(1,1,0)$	$(2,0,0)$
vs $(0,1,1)$	**1**	0	-1	-1	-2	-4

  Best: $(0,0,2)$ with 1 (base 2 lost, base 3 won: $-2+3$). New.

- *Baloney's oracle* (smallest payoff = best for Baloney). Best reply to $(0,1,1)$:

Baloney	$(0,0,2)$	$(0,1,1)$	$(0,2,0)$	$(1,0,1)$	$(1,1,0)$	$(2,0,0)$
vs $(0,1,1)$	$-$**1**	0	1	1	2	4

  Best: $(0,0,2)$ with -1. New.

- *Decision.* Both best responses are new: add $(0,0,2)$ to both sets.

Iteration 3.

- *We have* $T_1 = \{(1,1,0),(0,1,1),(0,0,2)\}$ and $T_2 = \{(1,1,0),(0,1,1),(0,0,2)\}$.
- *Restricted game.* $\begin{pmatrix} 0 & -2 & 0 \\ 2 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}$ (order $(1,1,0),(0,1,1),(0,0,2)$). Row minima $-2,-1,\mathbf{0}$, column maxima $2,1,\mathbf{0}$: saddle point $(0,0,2)$. Value 0.
- *Blotto's oracle* (largest payoff = best for Blotto). Best reply to $(0,0,2)$:

Blotto	$(0,0,2)$	$(0,1,1)$	$(0,2,0)$	$(1,0,1)$	$(1,1,0)$	$(2,0,0)$
vs $(0,0,2)$	$\mathbf{0}$	-1	-1	-2	$\mathbf{0}$	-2

  Best: 0, at $(0,0,2)$ and $(1,1,0)$, both already in T_1.
- *Baloney's oracle* (smallest payoff = best for Baloney). Best reply to $(0,0,2)$:

Baloney	$(0,0,2)$	$(0,1,1)$	$(0,2,0)$	$(1,0,1)$	$(1,1,0)$	$(2,0,0)$
vs $(0,0,2)$	$\mathbf{0}$	1	1	2	$\mathbf{0}$	2

  Best: 0, at $(0,0,2)$ and $(1,1,0)$, both already in T_2.
- *Decision.* Both best responses are already in their sets: **stop**. Equilibrium $(0,0,2)$ for both, value 0.

iter.	$T_1 = T_2$	added
1	$\{(1,1,0)\}$	$(0,1,1)$
2	$\{(1,1,0),(0,1,1)\}$	$(0,0,2)$
3	$\{(1,1,0),(0,1,1),(0,0,2)\}$	stop

The restricted value was 0 in every iteration, so it does not tell us when to stop; the membership test does.

(b) Now the two oracles answer differently.

Iteration 1.

- *We have* $T_1 = \{(1,1,0)\}$ and $T_2 = \{(0,1,1)\}$.
- *Restricted game.* 1×1: Blotto plays $(1,1,0)$, Baloney plays $(0,1,1)$. Value -2.
- *Blotto's oracle* (largest payoff = best for Blotto). Best reply to Baloney's $(0,1,1)$:

Blotto	$(0,0,2)$	$(0,1,1)$	$(0,2,0)$	$(1,0,1)$	$(1,1,0)$	$(2,0,0)$
vs $(0,1,1)$	$\mathbf{1}$	0	-1	-1	-2	-4

  Best: $(0,0,2)$ with 1. **New**.

- *Baloney's oracle* (smallest payoff = best for Baloney). Best reply to Blotto's $(1,1,0)$:

Baloney	$(0,0,2)$	$(0,1,1)$	$(0,2,0)$	$(1,0,1)$	$(1,1,0)$	$(2,0,0)$
vs $(1,1,0)$	0	$\mathbf{-2}$	-1	-1	0	1

  Best: $(0,1,1)$ with -2. **Already in** T_2.

- *Decision.* Baloney's best response is in T_2, Blotto's is not: add $(0,0,2)$ to T_1 only.

Iteration 2.

- *We have* $T_1 = \{(1,1,0),(0,0,2)\}$ and $T_2 = \{(0,1,1)\}$.
- *Restricted game.* 2×1: against $(0,1,1)$ Blotto's rows $(1,1,0),(0,0,2)$ give -2 and 1, so Blotto plays $(0,0,2)$ and Baloney plays $(0,1,1)$. Value 1.
- *Blotto's oracle* (largest payoff = best for Blotto). Best reply to Baloney's $(0,1,1)$: the same table as in iteration 1, best $(0,0,2)$ with 1. **Already in** T_1.
- *Baloney's oracle* (smallest payoff = best for Baloney). Best reply to Blotto's $(0,0,2)$:

Baloney	$(0,0,2)$	$(0,1,1)$	$(0,2,0)$	$(1,0,1)$	$(1,1,0)$	$(2,0,0)$
vs $(0,0,2)$	$\mathbf{0}$	1	1	2	$\mathbf{0}$	2

  Best: 0, at $(0,0,2)$ and $(1,1,0)$ (a tie); both are **new**. We take $(0,0,2)$.

- *Decision.* Now Blotto's best response is in T_1, Baloney's is not: add $(0,0,2)$ to T_2 only.

Iteration 3.

- *We have* $T_1 = \{(1,1,0),(0,0,2)\}$ and $T_2 = \{(0,1,1),(0,0,2)\}$.
- *Restricted game.* $\begin{pmatrix} -2 & 0 \\ 1 & 0 \end{pmatrix}$ (rows $(1,1,0),(0,0,2)$; columns $(0,1,1),(0,0,2)$). Saddle point $((0,0,2),(0,0,2))$. Value 0.
- *Blotto's oracle* (largest payoff = best for Blotto). Best reply to $(0,0,2)$: best 0 at $(0,0,2)$ and $(1,1,0)$, both **already in** T_1.
- *Baloney's oracle* (smallest payoff = best for Baloney). Best reply to $(0,0,2)$: best 0 at $(0,0,2)$, **already in** T_2. (It ties with $(1,1,0) \notin T_2$, but a tie gains nothing.)
- *Decision.* Both best responses are in their sets: **stop**.

iter.	T_1	T_2	restricted value	added
1	$(1,1,0)$	$(0,1,1)$	-2	$(0,0,2)$ to T_1
2	$(1,1,0),(0,0,2)$	$(0,1,1)$	1	$(0,0,2)$ to T_2
3	$(1,1,0),(0,0,2)$	$(0,1,1),(0,0,2)$	0	stop

What (b) shows:

- The algorithm stops only when *both* best responses are already in their sets. In iteration 1 Baloney was satisfied but Blotto was not; one satisfied player is not enough.
- Each player's set grows on its own, so the restricted game need not be square (2×1 in iteration 2).
- The restricted value jumps $-2 \to 1 \to 0$; it is not a stopping signal.
- *Why the test works.* If Blotto's best response lies in T_1, no allocation of the full game does better against Baloney's restricted strategy than what Blotto already plays. If the same holds for Baloney, neither player can improve with any strategy at all: the restricted equilibrium is an equilibrium of the full game. (A best response that only *ties* with one inside the set gains nothing, as in iteration 3.)

Tutorial 5: Alternatives to Nash equilibrium definitions

Correlated equilibrium (CE). A mediator draws a profile $s \in S$ from a known distribution p and tells each player *privately* only his own recommendation s_i. p is a correlated equilibrium if for every player i and all $s_i, t_i \in S_i$

$$\underbrace{\sum_{s_{-i} \in S_{-i}} p(s_i, s_{-i})\, u_i(s_i, s_{-i})}_{\text{obey } s_i} \;\geq\; \underbrace{\sum_{s_{-i} \in S_{-i}} p(s_i, s_{-i})\, u_i(t_i, s_{-i})}_{\text{deviate to } t_i}.$$

Theorem. If $(p_1, \ldots, p_n)$ is a Nash equilibrium, then $p(s_1, \ldots, s_n) = \prod_{i \in N} p_i(s_i)$ is a correlated equilibrium.

Stackelberg game. The *leader* (player 1) commits to a mixed strategy $p_1 \in \Delta_1$. The *follower* (player 2) observes p_1, not the realized action, and plays a best response

$$s_2 \in \mathrm{BR}_2(p_1) = \arg\max_{t_2 \in S_2} U_2(p_1, t_2).$$

The leader chooses p_1 to *maximize her own payoff*, anticipating this response.

Strong and weak Stackelberg equilibrium. If $\mathrm{BR}_2(p_1)$ has several elements, ties are broken for the leader (strong) or against her (weak). A pair (p_1^*, s_2^*) with $s_2^* \in \mathrm{BR}_2(p_1^*)$ is

a *strong Stackelberg equilibrium* (SSE) if $\quad p_1^* \in \arg\max_{p_1 \in \Delta_1} \max_{s_2 \in \mathrm{BR}_2(p_1)} U_1(p_1, s_2)$,

$s_2^* \in \arg\max_{s_2 \in \mathrm{BR}_2(p_1^*)} U_1(p_1^*, s_2)$,

a *weak Stackelberg equilibrium* (WSE) if $\quad p_1^* \in \arg\max_{p_1 \in \Delta_1} \min_{s_2 \in \mathrm{BR}_2(p_1)} U_1(p_1, s_2)$,

$s_2^* \in \arg\min_{s_2 \in \mathrm{BR}_2(p_1^*)} U_1(p_1^*, s_2)$.

Three optimizations: the follower maximizes U_2 (this gives BR_2), the tie is broken by max or min of U_1 over BR_2, and the leader maximizes over p_1. An SSE always exists; a WSE may not (the outer maximum need not be attained).

Computing an SSE. For every follower action $s_2 \in S_2$ solve the linear program

$$\max_{p_1 \in \Delta_1} U_1(p_1, s_2) \qquad \text{subject to} \qquad U_2(p_1, s_2) \;\geq\; U_2(p_1, t_2) \quad \text{for all } t_2 \in S_2,$$

and take the best feasible one.

5 ALTERNATIVES TO NASH EQUILIBRIUM

Exercise 5.24
The concept of correlated equilibrium is a generalization of Nash equilibrium in the following sense. Let $G = (N, (S_i)_{i\in N}, (u_i)_{i\in N})$ be an n-player strategic game, $(p_1, \ldots, p_n)$ be its Nash equilibrium in mixed strategies, and define

$$p(s_1, \ldots, s_n) := \prod_{i\in N} p_i(s_i), \qquad (s_1, \ldots, s_n) \in S = S_1 \times \cdots \times S_n.$$

Then p is a correlated equilibrium of G. Prove this claim.

Exercise 5.25
Imagine a situation, where 2 drivers are driving the same road in the opposite direction. First driver is in the correct lane, while the second one is in the opposite. If they both continue in the same lane, they will crash into each other, which results in the reward 0, similarly if they both change the lane. However, if only one of them switches lane they will miss each other, which is beneficial for both of them and they will receive reward 3. Moreover, if the driver is in the wrong lane, it will be penalized with reward -1, regardless of the outcome of the game. Model this as a normal-form game and find all of it's correlated equilibria.

(a) Model this as a normal-form game.

(b) Write the conditions of correlated equilibrium as linear inequalities in the probabilities of the four strategy profiles and describe the set of all correlated equilibria.

(c) Find the mixed Nash equilibrium and check that the distribution it induces satisfies the inequalities from (b).

(d) A traffic light tells, with probability $\frac{1}{2}$ each, "Driver 1 changes" or "Driver 2 changes". Is this a correlated equilibrium? Can it be obtained from a Nash equilibrium as in the previous exercise?

Exercise 5.26
Show that there exists a unique correlated equilibrium in the following game

		Player 2	
		L	R
Player 1	T	$1, 0$	$c, 1+d$
	B	$0, 1$	$1+a, b$

in which $a, b, c, d \in (-\frac{1}{4}, \frac{1}{4})$. Find this correlated equilibrium. What is the limit of the correlated equilibrium payoff as a, b, c, d approach 0?

Exercise 5.27

Nash, moving first, committing. In a *Stackelberg game*, the row player (the *leader*) publicly commits to a strategy, possibly mixed. The column player (the *follower*) observes the commitment, but not the outcome of the randomization, and plays a best response. Consider the game

Leader \ Follower	L	R
T	$4,1$	$6,0$
B	$2,0$	$5,3$

(a) *Simultaneous moves.* Find all Nash equilibria and the leader's payoff.

(b) *Sequential moves.* The leader moves first and the follower sees her *action* before choosing. Draw the game tree and find the leader's payoff by backward induction.

(c) *Commitment.* The leader commits to a mixed strategy p_1; write $p = p_1(B)$ for the probability that she plays B. The follower sees only p. For which p is R a best response? Which p is best for the leader, if the follower breaks ties in the leader's favour? Solve it by one linear program for each follower action.

(d) What if, when indifferent, the follower plays L instead (ties broken against the leader), i.e., Weak Stacklberg equilibrium?

(e) Compare the payoffs of both players in (a)-(c). Why can the leader get more in (c) than in (b)?

Exercise 5.28

The row player is the leader and publicly commits to a mixed strategy p_1; write $p = p_1(B)$ for the probability that she plays B:

L \ F	a	b	c
T	$1,3$	$0,2$	$4,0$
B	$0,0$	$3,2$	$1,4$

(a) Find the pure Nash equilibrium. Then let the leader move first, with the follower seeing her *action* (a sequential game). What does the leader get?

(b) For each follower action $j \in \{a, b, c\}$, write the linear program: maximize the leader's expected utility over $p \in [0, 1]$, subject to j being a best response of the follower. Solve each of the three programs.

(c) Find the strong Stackelberg equilibrium and the leader's payoff. Compare with both versions in (a).

(d) What happens if the follower breaks ties against the leader? Does a weak Stackelberg equilibrium exist?

Exercise 5.29

Consider following utility matrix two-person normal-form game, where row player is a leader, which publicly announces its strategy

		F				
		a	*b*	*c*	*d*	*e*
L	*T*	2, 4	6, 4	9, 0	1, 2	7, 4
	B	8, 4	0, 4	3, 6	1, 5	0, 0

Find Strong and Weak Stackelberg Equilibrium

SOLUTIONS

Solution 5.24

Clearly, p is a probability distribution on the joint strategy space S by the definition. We know that p is a correlated equilibrium if the inequality

$$\sum_{s_{-i} \in S_{-i}} p(s_i, s_{-i}) \cdot u_i(s_i, s_{-i}) \geq \sum_{s_{-i} \in S_{-i}} p(s_i, s_{-i}) \cdot u_i(s'_i, s_{-i}). \tag{19}$$

holds for each player i and all strategies $s_i, s'_i \in S_i$. Since $(p_1, \ldots, p_n)$ is a Nash equilibrium, it follows that for each player $i \in N$ and all strategies $s_i, s'_i \in S_i$ such that $p_i(s_i) > 0$,

$$U_i(s_i, p_{-i}) \geq U_i(s'_i, p_{-i}), \tag{20}$$

where U_i denotes the expected utility (payoff) of player i. Since

$$p_i(s_i) \cdot U_i(s_i, p_{-i}) = \sum_{s_{-i} \in S_{-i}} p_i(s_i) \cdot \prod_{j \neq i} p_j(s_j) \cdot u_i(s_i, s_{-i}) = \sum_{s_{-i} \in S_{-i}} p(s_i, s_{-i}) \cdot u_i(s_i, s_{-i})$$

and

$$p_i(s_i) \cdot U_i(s'_i, p_{-i}) = \sum_{s_{-i} \in S_{-i}} p_i(s_i) \cdot \prod_{j \neq i} p_j(s_j) \cdot u_i(s'_i, s_{-i}) = \sum_{s_{-i} \in S_{-i}} p(s_i, s_{-i}) \cdot u_i(s'_i, s_{-i}),$$

from (20) we get (19). In case that $p_i(s_i) = 0$, necessarily $p(s_i, s_{-i}) = 0$ for every $s_{-i} \in S_{-i}$, so that boths sides of (19) are zero.

Solution 5.25

(a) The normal-form representation of this game is

		Driver 2	
		Stay	*Change*
Driver 1	*Stay*	$0, -1$	$3, 3$
	Change	$2, 2$	$-1, 0$

(b) Now let us denote all of the 4 possible joint strategies followingly

$$\alpha = p(Stay, Stay) \tag{21}$$
$$\beta = p(Stay, Change) \tag{22}$$
$$\gamma = p(Change, Stay) \tag{23}$$
$$\delta = p(Change, Change) \tag{24}$$

The inequalities for the Correlated equilibrium are then

$$0\alpha + 3\beta \geq 2\alpha - 1\beta \tag{25}$$
$$2\gamma - 1\delta \geq 0\gamma + 3\delta \tag{26}$$
$$-1\alpha + 2\gamma \geq 3\alpha + 0\gamma \tag{27}$$
$$3\beta + 0\delta \geq -1\beta + 2\delta \tag{28}$$

This may be simplified to

$$2\beta \geq \alpha \tag{29}$$
$$\gamma \geq 2\delta \tag{30}$$
$$\gamma \geq 2\alpha \tag{31}$$
$$2\beta \geq \delta \tag{32}$$

Some interesting correlated equilibria are that if $\alpha = \delta = 0$, then any mixed strategy with support β, γ is a Correlated equilibrium. If $\delta = 0$, then $\alpha \in [0, \frac{2}{7}]$, similarly if $\alpha = 0$, then $\delta \in [0, \frac{2}{7}]$

(c) Let x and y be the probabilities of *Stay* for Driver 1 and Driver 2. Driver 2 is indifferent when $-x + 2(1-x) = 3x$, i.e. $x = \frac{1}{3}$. Driver 1 is indifferent when $3(1-y) = 2y - (1-y)$, i.e. $y = \frac{2}{3}$. The induced distribution is

$$\alpha = xy = \tfrac{2}{9}, \quad \beta = x(1-y) = \tfrac{1}{9}, \quad \gamma = (1-x)y = \tfrac{4}{9}, \quad \delta = (1-x)(1-y) = \tfrac{2}{9}.$$

All four inequalities from (b) hold, even with equality: $2\beta = \alpha$, $\gamma = 2\delta$, $\gamma = 2\alpha$, $2\beta = \delta$. (Equality is the indifference principle again: in a fully mixed equilibrium, every recommendation leaves the player indifferent.) The game also has the two pure Nash equilibria $(\mathit{Stay}, \mathit{Change})$ and $(\mathit{Change}, \mathit{Stay})$, i.e. $\beta = 1$ and $\gamma = 1$.

(d) Here $\alpha = \delta = 0$ and $\beta = \gamma = \frac{1}{2}$, so by (b) it is a correlated equilibrium: a driver who is told to change knows the other one stays, and vice versa. It is not a product of mixed strategies, because a product distribution satisfies $\alpha\delta = \beta\gamma$, while here $\alpha\delta = 0 \neq \frac{1}{4} = \beta\gamma$. So it is not induced by any Nash equilibrium. The same argument applies to $(\frac{2}{7}, \frac{1}{7}, \frac{4}{7}, 0)$ from (b), where $\alpha\delta = 0 \neq \frac{4}{49} = \beta\gamma$. Moreover, this one cannot even be obtained by randomly choosing one of the Nash equilibria: every mixture of the three Nash distributions has $\alpha = \delta$.

Solution 5.26

Let us start by writing all the correlated equilibrium conditions

$$\begin{aligned} p(T,L) + (c-1-a)p(T,R) &\geq 0 \\ -p(B,L) + (1+a-c)p(B,R) &\geq 0 \\ (-1-d)p(T,L) + (1-b)p(B,L) &\geq 0 \\ (1+d)p(T,R) + (b-1)p(B,R) &\geq 0 \end{aligned}$$

Now let us introduce following substitions

$$\begin{aligned} X &= 1 + a - c \\ Y &= 1 - b \\ Z &= 1 + d \end{aligned}$$

If we rewrite the conditions with following substitions we will end up with

$$\begin{aligned} p(T,L) &\geq Xp(T,R) \\ Xp(B,R) &\geq p(B,L) \\ Yp(B,L) &\geq Zp(T,L) \\ Zp(T,R) &\geq Yp(B,R) \end{aligned}$$

Since we know that each of $a, b, c, d \in (-\frac{1}{4}, \frac{1}{4})$, we can find the domain for our substitions

$$\begin{aligned} X &\in (\frac{1}{2}, \frac{3}{2}) \\ Y &\in (\frac{3}{4}, \frac{5}{4}) \\ Z &\in (\frac{3}{4}, \frac{5}{4}) \end{aligned}$$

This means that we can multiply or divide with them without changing the inequality. Now let us use this to get following inequalities. First we use the first inequality, than the fourth, then second and lastly the third

$$p(T,L) \geq Xp(T,R) \geq \frac{XY}{Z}p(B,R) \geq \frac{XY}{XZ}p(B,L) \geq \frac{XYZ}{XYZ}p(T,L) = p(T,L)$$

We see that in order for this to hold, all of the inequalities have to be equalities. This means that we can express each probability by $p(T,L)$

$$\begin{aligned} p(T,R) &= \frac{p(T,L)}{X} \\ p(B,R) &= \frac{Zp(T,L)}{XY} \\ p(B,L) &= \frac{Zp(T,L)}{Y} \end{aligned}$$

Now we can find the actual prbabilities from the condition that all of these has to sum up to 1

$$\begin{aligned} p(T,L) + \frac{p(T,L)}{X} + \frac{Zp(T,L)}{XY} + \frac{Zp(T,L)}{Y} &= 1 \\ (XY + Y + Z + ZX)p(T,L) &= XY \\ p(T,L) &= \frac{XY}{(XY + Y + Z + ZX)} \end{aligned}$$

Out of this we can get all other probabilities

$$\begin{aligned} p(T,R) &= \frac{Y}{XY + Y + Z + ZX} \\ p(B,R) &= \frac{Z}{XY + Y + Z + ZX} \\ p(B,L) &= \frac{ZX}{XY + Y + Z + ZX} \end{aligned}$$

The value of this equilibrium is than

$$\begin{aligned} u_1 &= p(T,L) + cp(T,R) + (1+a)p(B,R) = \frac{XY + cY + (1+a)Z}{XY + Y + Z + ZX} \\ u_2 &= p(B,L) + (1+d)p(T,R) + bp(B,R) = \frac{ZX + (1+d)Y + bZ}{XY + Y + Z + ZX} \end{aligned}$$

The only thing that remains is expressing X, Y, Z with the original variables. In order to compute what happens when all of the original variables a, b, c, d goes to 0, we will keep the substitions. In such a case

$$\begin{aligned} X &= 1 \\ Y &= 1 \\ Z &= 1 \end{aligned}$$

In this case the resulting values are

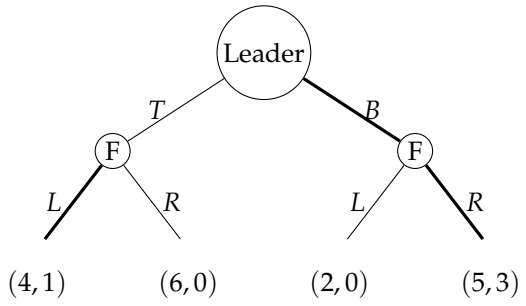
$$u_1 = 0.5$$
$$u_2 = 0.5$$

, which is the same as Nash equilibrium. If looking closely, one would notice that this game is symmetrical and is similair to the Matching Pennies from exercise **??**.

Solution 5.27

(a) T strictly dominates B for the leader ($4 > 2$ and $6 > 5$), and the follower's best response to T is L ($1 > 0$). The unique Nash equilibrium is (T, L) with payoffs $(4, 1)$.

(b) The follower sees the action, so he best-responds to each action separately:



After T the follower plays L (leader gets 4), after B he plays R (leader gets 5). Backward induction: the leader plays B, and the payoffs are $(5, 3)$. Moving first openly lets the leader use the dominated action B: the follower's response depends on it. Games of this kind, where later players observe earlier moves, are the topic of extensive-form games.

(c) The leader plays B with probability p and T with probability $1 - p$.

Step 1: the follower's expected utilities. Against the commitment p,

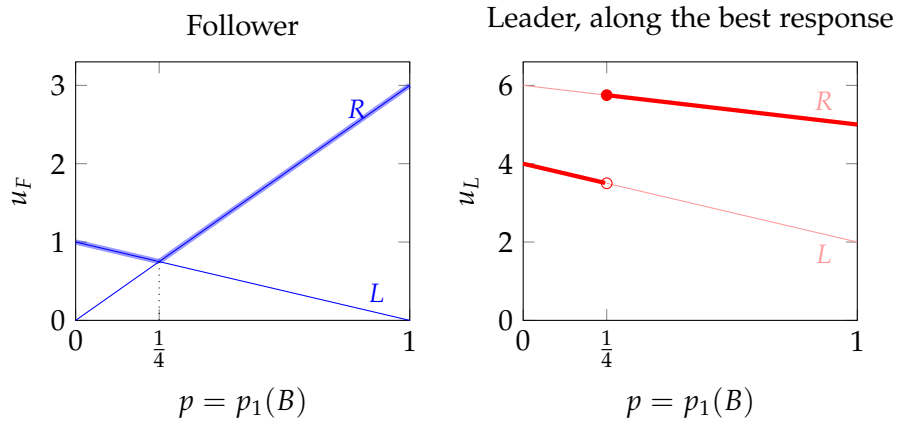
$$u_F(p, L) = 1 \cdot (1 - p) + 0 \cdot p = 1 - p, \qquad u_F(p, R) = 0 \cdot (1 - p) + 3 \cdot p = 3p.$$

L is strictly better iff $1 - p > 3p$, i.e. $p < \frac{1}{4}$; R is strictly better iff $p > \frac{1}{4}$; at $p = \frac{1}{4}$ both give $\frac{3}{4}$ and the follower is indifferent. So

$$\mathrm{BR}_F(p) = \begin{cases} \{L\}, & p < \frac{1}{4}, \\ \{L, R\}, & p = \frac{1}{4}, \\ \{R\}, & p > \frac{1}{4}. \end{cases}$$

Step 2: the leader's expected utilities against each follower action:

$$u_L(p, L) = 4(1 - p) + 2p = 4 - 2p, \qquad u_L(p, R) = 6(1 - p) + 5p = 6 - p.$$



Step 3: what the leader gets, with ties broken for her. Putting Steps 1 and 2 together,

$$\max_{s_2 \in \mathrm{BR}_2(p)} u_L(p, s_2) = \begin{cases} 4 - 2p, & p < \frac{1}{4} \quad \text{(only } L\text{)}, \\ \max\{4 - \frac{1}{2},\ 6 - \frac{1}{4}\} = \max\{\frac{7}{2}, \frac{23}{4}\} = \frac{23}{4}, & p = \frac{1}{4} \quad \text{(tie, take } R\text{)}, \\ 6 - p, & p > \frac{1}{4} \quad \text{(only } R\text{)}. \end{cases}$$

Step 4: maximize u_F over p. The *strong Stackelberg equilibrium*: the leader commits to T with probability $\frac{3}{4}$ and B with probability $\frac{1}{4}$, the follower plays R, and the expected utilities are $(\frac{23}{4}, \frac{3}{4})$.

The same answer can be done from linear programs. Fix the follower's action, require that it is a best response, and maximize the leader's payoff:

$$\mathrm{LP}_L: \quad \max\ 4 - 2p \quad \text{s.t.} \quad 1 - p \ \geq\ 3p, \ \ 0 \leq p \leq 1 \quad \Longrightarrow \quad p \leq \tfrac{1}{4}: \quad p = 0, \text{ value } 4,$$
$$\mathrm{LP}_R: \quad \max\ 6 - p \quad \text{s.t.} \quad 3p \ \geq\ 1 - p, \ \ 0 \leq p \leq 1 \quad \Longrightarrow \quad p \geq \tfrac{1}{4}: \quad p = \tfrac{1}{4}, \text{ value } \tfrac{23}{4}.$$

Each program has one variable, so its feasible set is an interval and the linear objective is maximized at an endpoint. The better program is LP_R, so the follower plays R and $p^* = \frac{1}{4}$. The constraint $3p \geq 1 - p$ allows equality, so the tie at $p = \frac{1}{4}$ is automatically broken for the leader. One program per follower action, then take the best.

(d) Steps 1 and 2 do not change: the follower's best responses and the leader's two lines in the figure are the same. Only the tie at $p = \frac{1}{4}$ is now broken *against* the leader.

Step 3(WSE): what the leader gets, with ties broken against her.

$$F_{\mathrm{W}}(p) = \min_{s_2 \in \mathrm{BR}_2(p)} u_L(p, s_2) = \begin{cases} 4 - 2p, & p < \frac{1}{4} \quad \text{(only } L\text{)}, \\ \min\{\frac{7}{2}, \frac{23}{4}\} = \frac{7}{2}, & p = \frac{1}{4} \quad \text{(tie, take } L\text{)}, \\ 6 - p, & p > \frac{1}{4} \quad \text{(only } R\text{)}. \end{cases}$$

The only difference from F_{S} is the single point $p = \frac{1}{4}$: the value there is $\frac{7}{2}$ instead of $\frac{23}{4}$.

Step 4(WSE): try to maximize over p. A weak Stackelberg equilibrium needs a p with the largest value of $F_{\mathrm{W}}(p)$. We show that every p is beaten by some other p':

- If $p < \frac{1}{4}$: $F_{\mathrm{W}}(p) = 4 - 2p \leq 4$, but $F_{\mathrm{W}}(0.3) = 6 - 0.3 = 5.7 > 4$.

- If $p = \frac{1}{4}$: $F_W(\frac{1}{4}) = \frac{7}{2} < 5.7 = F_W(0.3)$.
- If $p > \frac{1}{4}$: take the midpoint $p' = \frac{1}{2}(p + \frac{1}{4})$. Then $\frac{1}{4} < p' < p$, the follower still plays R, and $F_W(p') = 6 - p' > 6 - p = F_W(p)$.

Hence F_W has no maximum and the *weak Stackelberg equilibrium does not exist*.

(e)

	leader	follower
(a) simultaneous (Nash)	4	1
(b) sequential, action observed	5	3
(c) commitment, only p observed	$\frac{23}{4}$	$\frac{3}{4}$

Solution 5.28

(a) (T, a) is a Nash equilibrium with leader's payoff 1. In the sequential game the follower sees the action: after T he plays a (leader gets 1), after B he plays c (leader gets 1). So moving first openly gives the leader only 1 the same as committing to a pure strategy.

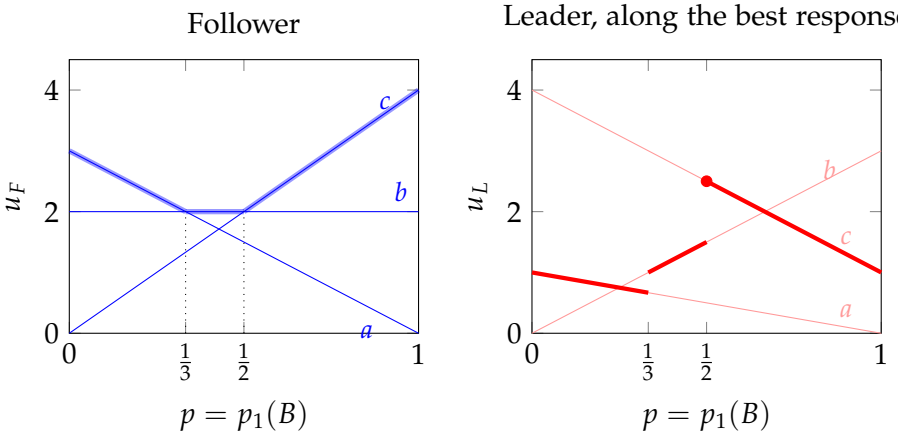
(b) The expected utilities are

	a	b	c
u_F	$3 - 3p$	2	$4p$
u_L	$1 - p$	$3p$	$4 - 3p$

For each follower action, the constraints say that this action is at least as good for the follower as the two others.

	maximize	subject to $(0 \le p \le 1)$	feasible p	optimum
LP_a	$1 - p$	$3 - 3p \ge 2,\ \ 3 - 3p \ge 4p$	$p \le \frac{1}{3}$	$p = 0$, value 1
LP_b	$3p$	$2 \ge 3 - 3p,\ \ 2 \ge 4p$	$\frac{1}{3} \le p \le \frac{1}{2}$	$p = \frac{1}{2}$, value $\frac{3}{2}$
LP_c	$4 - 3p$	$4p \ge 3 - 3p,\ \ 4p \ge 2$	$p \ge \frac{1}{2}$	$p = \frac{1}{2}$, value $\frac{5}{2}$

(c) The best of the three programs is LP_c. The strong Stackelberg equilibrium is: the leader commits to $(\frac{1}{2}, \frac{1}{2})$, the follower plays c, and the payoffs are $(\frac{5}{2}, 2)$. The leader gets $\frac{5}{2}$ instead of 1 from moving first openly.



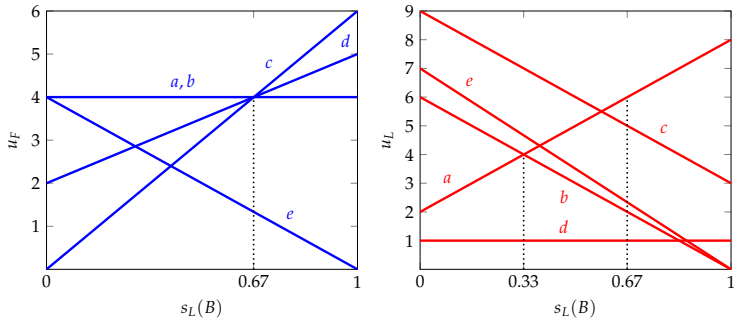
The follower plays a on $[0, \frac{1}{3}]$, b on $[\frac{1}{3}, \frac{1}{2}]$ and c on $[\frac{1}{2}, 1]$. Along this best response the leader's utility is highest at $p = \frac{1}{2}$ with c, which gives $\frac{5}{2}$.

(d) Only the ties change. At $p = \frac{1}{2}$ the follower now plays b instead of c, so the leader gets $\frac{3}{2}$ instead of $\frac{5}{2}$; at $p = \frac{1}{3}$ he plays a. For $p > \frac{1}{2}$ he still plays

c and the leader gets $4 - 3p$: e.g. 2.2, 2.47, 2.497 for $p = 0.6,\ 0.51,\ 0.501$ — closer and closer to $\frac{5}{2}$, never reaching it, while every $p \leq \frac{1}{2}$ gives at most $\frac{3}{2}$. So no p is best and the *weak Stackelberg equilibrium does not exist*, exactly as in the warm-up. Committing to $p = \frac{1}{2} + \varepsilon$ gives $\frac{5}{2} - 3\varepsilon$ under either rule.

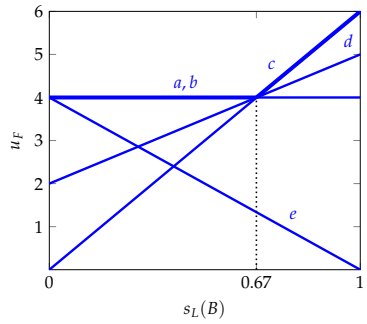
Solution 5.29

Since the leader has only two actions, we may visualize the expected utility based on the policy announced by the leader.



When computing the Stackelberg equilibrium, the follower always takes a best response to the leader's policy. Best responses are as follows

$$\mathrm{BR}(s_L) = \begin{cases} a, b, e & s_L(B) = 0, \\ a, b & 0 < s_L(B) < \frac{2}{3}, \\ a, b, c, d & s_L(b) = \frac{2}{3}, \\ c & \frac{2}{3} < s_L(b) \leq 1, \end{cases}$$



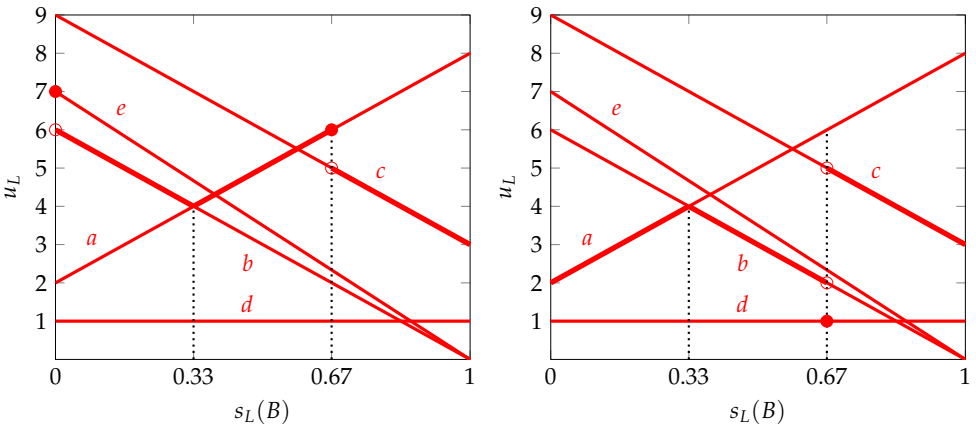
Now we know, based on the policy of the leader, what are the follower best responses. Now we have to figure out which best response the follower would take.

Let us define functions $\overline{\beta}(s_L)$ and $\underline{\beta}(s_L)$, which, based on the leader's policy, give pure action that the follower should play to maximize or minimize the leader's utility. Similarly we define function $\overline{\beta}(s_L)$ and $\underline{\beta}(s_L)$ that corresponds to the expected utility for leader when it fixes its policy.

$$\overline{\beta}(s_L) = \operatorname*{arg\,max}_{a_F \in BR(s_L)} u_L(s_L, a_F) \qquad \underline{\beta}(s_L) = \operatorname*{arg\,min}_{a_F \in BR(s_L)} u_L(s_L, a_F)$$

The policies are then

$$\overline{\beta}(s_L) = \begin{cases} e & s_L(B) = 0, \\ b & 0 < s_L(B) \leq \frac{1}{3}, \\ a & \frac{1}{3} < s_L(B) \leq \frac{2}{3}, \\ c & \frac{2}{3} < s_L(B) \leq 1, \end{cases} \qquad \underline{\beta}(s_L) = \begin{cases} a & 0 \leq s_L(B) < \frac{1}{3}, \\ b & \frac{1}{3} \leq s_L(B) < \frac{2}{3}, \\ d & s_L(B) = \frac{2}{3}, \\ c & \frac{2}{3} < s_L(B) \leq 1, \end{cases}$$



Strong Stackelberg equilibrium from the left plot is strategy $s_L(B) = 0$ because this maximizes the leader's utility. Weak Stackelberg equilibrium does not exist, because if we take $\varepsilon > 0$ and policy $s_L(B) = \frac{2}{3} + \varepsilon$. When lowering the value of ε, we always get better utility, and when $\varepsilon = 0$, the value of action c is 5, which is the best we can get. But when $\varepsilon = 0$, the utility drops to 1, because optimal action changes to d. Therefore we cannot set such ε, which would maximize the value, so the Weak Stackelberg equilibrium does not exist.

6 LEARNING IN GAMES

7 LEARNING IN GAMES

Exercise 7.30

Two roads. Three drivers travel from X to Y, each choosing road A or road B. If k drivers use road A, each of them needs k hours; if k drivers use road B, each of them needs $2k$ hours. Every driver wants to minimize his own travel time. The drivers adjust *one at a time*: in each step, one driver who can shorten his travel time switches to the other road; no two drivers switch simultaneously.

(a) Find all pure Nash equilibria. (It is enough to say how many drivers use road A.)

(b) Choose any starting situation and run the process. Does it stop? Where? What could go wrong if all drivers who want to switch did so at the same time?

(c) Why does it stop? Imagine the drivers arrive one by one and add up the time each of them needs at the moment he arrives:

$$\Phi(k_A, k_B) = \underbrace{(1 + 2 + \cdots + k_A)}_{\text{road } A} + \underbrace{(2 + 4 + \cdots + 2k_B)}_{\text{road } B}.$$

Compute Φ for all situations. Then show: whenever *one* driver switches roads, Φ changes by exactly the change of *his own* travel time.

(d) Why must the process stop, and why at a Nash equilibrium? Did your argument in (c) use the numbers k and $2k$?

Exercise 7.31

Going round in circles. Players switch one at a time to a best response: they alternate, and a player who cannot improve passes. A useful picture: in the payoff table, draw an arrow from each profile to the profile the moving player switches to.

(a) Come up with a 2×2 game in which this process never stops, whatever the starting profile. How do the arrows look? What must such a game lack?

(b) Come up with a game that *has* a pure Nash equilibrium, but in which the process reaches it from some starting profiles and cycles for ever from others. Compare with the two-roads game. *Hint:* start from your game in (a) and give each player a third action.

Exercise 7.32

Fictitious play in Matching Pennies. In *fictitious play* each player keeps track of how often the opponent has played each action so far and, in every round, plays a best response to these empirical frequencies. Both players choose at the same time, each using only the counts from the previous rounds. In Matching Pennies Player 1 (the matcher) gets 1 if the coins match and -1 otherwise; Player 2 gets the opposite. Start with (H, H) in round 1 and break ties in favour of H.

(a) Write down when Player 1 plays H and when Player 2 plays H, in terms of the counts. Then compute rounds 2 to 7.

(b) Predict: Will the *actions* converge? Will the *frequencies* converge, and to what?

(c) Run `fppennies`. Besides the frequencies α, β of H (of Player 1 and Player 2), it prints the lecture's saddle-point gap $g = |2\beta - 1| + |2\alpha - 1|$. What does g measure, and what happens to it?

Exercise 7.33

When frequencies are not enough. Fictitious play again. In the *anti-coordination game* both players get 1 if they choose *different* actions and 0 otherwise:

	A	B
A	$0, 0$	$1, 1$
B	$1, 1$	$0, 0$

Start with (A, A) in round 1 and break ties in favour of A.

(a) Compute rounds 2 to 6.

(b) What do the frequencies converge to? What do the players get per round, and what would they get by always playing A?

(c) Poll. Shapley's game:

	1	2	3
1	$1, 0$	$0, 1$	$0, 0$
2	$0, 0$	$1, 0$	$0, 1$
3	$0, 1$	$0, 0$	$1, 0$

(the row player gets 1 for the same number, the column player for one step ahead; the only Nash equilibrium is $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$ for both). After 30 000 rounds of fictitious play, are the row player's frequencies (i) close to $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$, (ii) close to a pure strategy, or (iii) still swinging? Run `fpshapley`.

Exercise 7.34

Predictable is exploitable. Every day you choose one of two actions A and B. Afterwards you learn the payoffs $u^t(A), u^t(B) \in [0,1]$ of *both* actions. Think of $u^t(a) = u(a, s^t_{-i})$ as the payoff of a against whatever your opponent did on day t. Let G_A, G_B be the total payoffs of A and B over the days so far. The *regret* of action a is $R(a) = G_a -$ your total payoff, and after T days your regret is

$$\mathrm{Reg}(T) = \max_{a \in \{A,B\}} [R(a)]_+,$$

i.e. how much worse you did than the best *fixed* action in hindsight. Consider the payoffs

$$(0,1),\ (1,0),\ (0,1),\ (1,0),\ (0,1),\ (1,0),\ (0,1),\ (1,0) \qquad (\text{day } 1,\dots,8).$$

(a) *Follow the leader* (FTL): every day choose the action with the largest total payoff so far (ties: A). This is fictitious play against the payoffs: a best response to the history. Compute FTL's total payoff, G_A, G_B, and the regret.

(b) What must a good rule do? Why do we compare with the best *fixed* action and not with the best action of each day?

Exercise 7.35

Regret matching by hand. Every day you choose one of two actions A and B and afterwards learn the payoffs of both, as before:

$$(0,1),\ (1,0),\ (0,1),\ (1,0),\ (0,1),\ (1,0),\ (0,1),\ (1,0) \qquad (\text{day } 1,\dots,8).$$

Regret matching chooses *probabilities*; we count your payoff in expectation. Keep two quantities apart:

- the *expected payoff of day t* (one day only): $p^t(A)\,u^t(A) + p^t(B)\,u^t(B)$;
- *your total payoff* (the whole history so far): the sum of the expected payoffs of all days so far. Similarly, G_A and G_B are the total payoffs of A and B so far.

The *regret* of action a so far is $R(a) = G_a -$ your total payoff: how much more you would have got by always playing a. *Regret matching* plays a on the next day with probability $\dfrac{[R(a)]_+}{[R(A)]_+ + [R(B)]_+}$, where $[x]_+ = \max(x, 0)$; if neither regret is positive, it plays $(\frac{1}{2}, \frac{1}{2})$. (The lecture counts the payoff of the action actually drawn; this is the same rule averaged over the coin.)

(a) Compute for days 1 to 4: the regrets before the day, $p^t(A)$, the expected payoff of the day, your total payoff, G_A, G_B, and the regrets after the day. Compare your regret after 4 days, $\mathrm{Reg} = \max\big([R(A)]_+, [R(B)]_+\big)$, with that of follow-the-leader.

(b) Predict: after 10 000 days, is the regret of regret matching (i) about 5000, like follow-the-leader, (ii) about 50, or (iii) about 0?

(c) Run `rmregret`. How does the regret grow with T?

Exercise 7.36

Learning against each other (predict, run, discuss). Now *both* players use regret matching against each other, each computing its regrets against the other's mixed strategy. We look at the current strategies and at the *average* strategies over all rounds so far.

(a) Matching Pennies, starting at (H, H). Predict: do the *current* strategies settle? Do the *averages*, and to what? Then run `rmpennies`.

(b) Discuss: why should the averages end up at the equilibrium? (Hint: what kind of game is Matching Pennies?)

(c) Restricted Rock-Paper-Scissors, where Bob may not play Scissors (payoffs to Alice):

	R	P
R	0	-1
P	1	0
S	-1	1

Predict: will the averages reproduce the LP solution of this game? Will Alice keep playing Rock? Then run `rmrps`.

(d) Discuss: when would you compute an equilibrium this way instead of with an LP?

SOLUTIONS

Solution 7.30

(a) $k_A = 2$: both roads take 2 hours; switching would give 4 (to B) or 3 (to A). For $k_A = 0, 1, 3$ some driver can improve.

(b) Every start ends at $k_A = 2$ (e.g. $0 \to 1 \to 2$, $3 \to 2$). If everybody switched at once: from $k_A = 0$ all move to A, get 3 hours, all move back, and so on.

(c) $\Phi = 12,\ 7,\ 5,\ 6$ for $k_A = 0, 1, 2, 3$. A driver leaving road X removes the *last* term of X's sum (his old time) and adds the *next* term of Y's sum (his new time), so for instance when such a player change from road B to the road A we get:

$$\Phi_{\text{new}} - \Phi_{\text{old}} = (k_A + 1) - 2k_B = \text{his new time} - \text{his old time}.$$

Such a Φ is a *potential* (in the lecture's utilities: $P = -\Phi$).

(d) Each improving switch lowers Φ strictly, and there are finitely many situations, so the process stops and it is where nobody can improve, i.e. at a Nash equilibrium. The argument in (c) never used k and $2k$, so it works for any road costs: every congestion game has a potential and a pure Nash equilibrium (Rosenthal).

Solution 7.31

(a) Matching Pennies: the arrows go round, $(H,H) \to (H,T) \to (T,T) \to (T,H) \to (H,H)$. To cycle from *every* start the game needs no pure Nash equilibrium: a cell with no arrow leaving it is one.

(b) For example

	x	y	z
a	$1,0$	$0,1$	$0,0$
b	$0,1$	$1,0$	$0,0$
c	$0,0$	$0,0$	$2,2$

(column moves first): from (a,x) it cycles in the upper left corner; from (c,x) it stops at (c,z). A pure Nash equilibrium alone does not make the process stop from every start; a potential does.

Solution 7.32

(a) With n_H^i, n_T^i the counts of Player i before the round: Player 1 plays H if $n_H^2 \geq n_T^2$ (copy), Player 2 plays H if $n_H^1 \leq n_T^1$ (oppose); ties to H. (Comparing counts is enough: the expected utilities are the counts divided by the same $t-1$.)

t	(n_H^2, n_T^2)	P1	(n_H^1, n_T^1)	P2	profile
1	$-$	H	$-$	H	HH
2	$(1,0)$	H	$(1,0)$	T	HT
3	$(1,1)$	H	$(2,0)$	T	HT
4	$(1,2)$	T	$(3,0)$	T	TT
5	$(1,3)$	T	$(3,1)$	T	TT
6	$(1,4)$	T	$(3,2)$	T	TT
7	$(1,5)$	T	$(3,3)$	H	TH

(b), (c) The actions keep cycling; the frequencies converge to $(\frac{1}{2}, \frac{1}{2})$, the Nash equilibrium. The gap g is what the players could gain by best-responding

to the frequencies (0 exactly at equilibrium); it falls from 0.40 (10 rounds) to 0.014 (10 000 rounds), not every round. Zero-sum games always behave like this (Robinson, 1951).

Solution 7.33

(a) Each player plays the opposite of the opponent's more frequent action; both see the same counts, so they always match: $(A,A),(B,B),(A,A),(B,B),(A,A),(B,B)$.

(b) The frequencies converge to $(\frac{1}{2},\frac{1}{2})$, the mixed Nash equilibrium — but the players get 0 every round, while always playing A would give $\frac{1}{2}$. The frequencies look right, the play is bad. Measuring this is what *regret* does. (From (A,B) they would stay at (A,B) and get 1.)

(c) (iii). The frequencies swing, e.g. $(0.50, 0.16, 0.34)$ after 10 000 rounds and $(0.25, 0.54, 0.21)$ after 30 000; the gap stays about 0.29. Not zero-sum, so fictitious play need not converge (Shapley, 1964).

Solution 7.34

(a) FTL plays $A,B,A,B,\ldots$ and always picks the action that gets 0: total 0, $G_A = G_B = 4$, regret $4 = T/2$.

(b) Not be predictable: the opponent foresaw FTL's choice and made it pay 0, and any deterministic rule can be fooled this way. So a good rule must *randomize*. The best action of each day is no fair benchmark: if a coin decides each day which action pays 1, every rule gets about $T/2$ while the per-day best gets T.

Solution 7.35

(a)

day	R before	$p^t(A)$	payoffs	exp. payoff	total	(G_A, G_B)	R after
1	$(0,0)$	$\frac{1}{2}$	$(0,1)$	$\frac{1}{2}$	$\frac{1}{2}$	$(0,1)$	$(-\frac{1}{2},\frac{1}{2})$
2	$(-\frac{1}{2},\frac{1}{2})$	0	$(1,0)$	0	$\frac{1}{2}$	$(1,1)$	$(\frac{1}{2},\frac{1}{2})$
3	$(\frac{1}{2},\frac{1}{2})$	$\frac{1}{2}$	$(0,1)$	$\frac{1}{2}$	1	$(1,2)$	$(0,1)$
4	$(0,1)$	0	$(1,0)$	0	1	$(2,2)$	$(1,1)$

After 4 days Reg $= 1$; follow-the-leader has 2.

(b), (c) (ii). Regret after $T = 8, 100, 1000, 10\,000$: follow-the-leader $4, 50, 500, 5000$; regret matching 1.45, 5.04, 15.83, 50.01. It grows like $\sqrt{T}$, so the regret *per day* goes to 0 — for every sequence of payoffs (Hart and Mas-Colell).

Solution 7.36

(a) The current strategies keep jumping; the averages approach $(\frac{1}{2},\frac{1}{2})$ and the payoff the value 0. What converges is the average, not the play.

(b) It is zero-sum: by the lecture's theorem the gap is at most $\rho_1 + \rho_2$, and regret matching drives both average regrets to 0. In general games this fails.

(c) Yes: after 10 000 rounds Alice $\approx$ (0, 0.67, 0.33), Bob $\approx$ (0.34, 0.66), the LP solution $(0, \frac{2}{3}, \frac{1}{3})$, $(\frac{1}{3}, \frac{2}{3})$. Rock (dominated) dies out.

(d) In games too large for an LP: rounds are cheap and each player needs only his own payoffs. Stop when $\rho_1 + \rho_2$ is small and output the averages — a certified approximate equilibrium. This is how CFR solves poker.

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